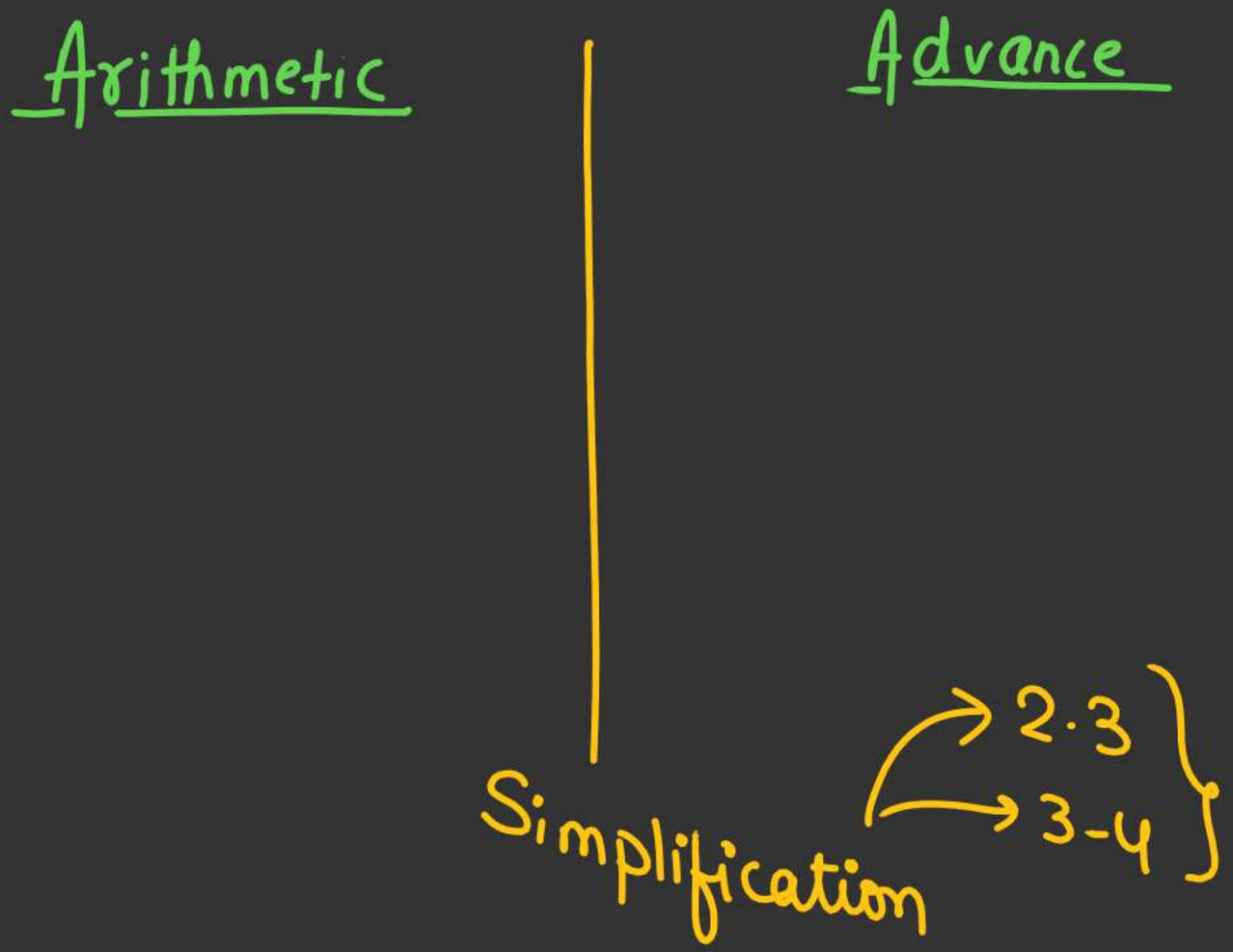
SIMPLIFICATION

सरलीकरण

CLASS NOTES


Arithmetic
Advance
Simplification
2.3
3-4

Type-1 BODMAS

$3+3-3\times3\div3$

$= 3+3-3\times1$

$= 3+3-3$

$= 3$

CONCEPT-01 (BODMAS Rule)

This rule is the basic principle of solving algebraic or numerical expressions.

यह नियम बीजगणितीय या संख्यात्मक व्यंजकों को हल करने का मूल सिद्धांत है।

V BODMAS

Order to solve/हल करने के क्रम

V → Vinicullum/Bar ( )

B → Bracket (कोष्ठक) { }

[ ]

O → of (का)

D → Division (भाग)

M → Multiplication (गुणा)

A → Addition (जोड़)

S → Subtraction (घटाव)

Types of Bracket & Solving order

कोष्ठक के प्रकार एवं हल करने के क्रम

(i) $\overline{\quad}$ → Vinculum/Line/Bar bracket (रेखा कोष्ठक)

(ii) ( ) → Small bracket (छोटा कोष्ठक)

(iii) { } → Curly bracket (मंझला कोष्ठक)

(iv) [ ] → Square bracket (बड़ा कोष्ठक)

BODMAS

1. The value of $9 \times 9 + 9 \div 9 - 9 \times 9 + 9 + 9 \times 9 - 9 - 9 \times 9$ is:

$9 \times 9 + 9 \div 9 - 9 \times 9 + 9 + 9 \times 9 - 9 - 9 \times 9$ का मान है:

SSC CPO 27/06/2024 (Shift-03)

(a) 9
(b) 1
(c) 81
(d) 0

$9 \times 9 + 1 - 9 \times 9 + 9 + 9 \times 9 - 9 - 9 \times 9$

$+81 + 1 - 81 + 9 + 81 - 9 - 81$

$= 1$

* $23 - 0.5 = 22.5$

* $18 - 0.7 = 17.3$

* $20 - 1.5 = 18.5$

* $19 - 2.37 = 16.63$

* $22 - 6.81 = 15.19$

2. Simplify the following:

निम्नलिखित का सरलीकरण करें:

$21 - 4.9 \div 7 + 3.9 \times 0.4 + 0.9$

SSC CGL 09/09/2024 (Shift-02)

(a) 18.23 **(b) 24.12**

(c) 26.32 **(d) 22.76**

$= 21 - 4.9 \times \frac{1}{7} + 3.9 \times 0.4 + 0.9$

$= 21 - 0.7 + 1.56 + 0.90$

$= 20.30 + 2.46$

$= 22.76$

BODMAS

3. **Simplify:/**निम्नलिखित को सरल कीजिए।

[25 – 48 ÷ 6 + 12 × 2] + 78 – [5 + 3 of (25 – 2 × 10)]

SSC CGL 25/09/2024 (Shift-02)

(a) 98 **(b) 90**

(c) 99 **(d) 89**

$[25-48\div6+12\times2]+78-[5+3\text{ of }(25-2\times10)]$

$=[25-8+24]+78-[5+3\text{ of }5]$

$=41+78-20$

$=119-20$

$=99$

BODMAS

4. Simplify

निम्नलिखित का मान ज्ञात करें।

$$2.5 \times [144 \div 198 \times \{121 \times 81 \div (11 \times 9)\}]$$

SSC CGL 17/07/2023 (Shift-01)

(a) 180 ✓
(b) 175
(c) 185
(d) 190

$$2.5 \times [144 \div 198 \times \{121 \times 81 \div 99\}]$$

$$= 2.5 \times \left[\frac{144}{198} \times 99\right]$$

$$= 2.5 \times 72$$

$$= 180$$

5. **Simplify./**निम्न का मान ज्ञात करें। 83 - 60

$325 + 276 \div [150 - \{9 \times 9 + (83 - 4 \times 15)\}]$

SSC CGL 20/07/2023 (Shift-03)

(a) 332 (b) 333

(c) 334 (d) 331

$325+276 \div [150 - \{9 \times 9 + 23\}]$

$= 325+276 \div [150 - 104]$

$= 325+276 \div 46$ (6)

$= 325+6$

$= 331$

$\Rightarrow 2-5+2P=3$

$\Rightarrow 2P=3$

6. If (48 ÷ 72 × 3) – [15 ÷ 8 × (40 – 32) – 10] + 2P = 6 ÷ 2, then find the value of P?

यदि (48 ÷ 72 × 3) – [15 ÷ 8 × (40 – 32) – 10] + 2P = 6 ÷ 2, तो P का मान ज्ञात कीजिए?

SSC CPO 03/10/2023 (Shift-3)

(a) 2 (b) 4

(c) 1 (d) 3

$\left(\frac{48}{72}\times 3\right)-[15\div 8\times 8-10]+2P=3$

$\Rightarrow 2-[15\times\frac{1}{8}\times 8-10]+2P=3$

$= \frac{1}{8} + \frac{5}{36} - \frac{36}{36} + \frac{27}{36}$

$= \frac{1}{8} - \frac{4}{36} \frac{1}{9}$

$= \frac{9-8}{72} = \frac{1}{72}$

7. What is the value of/निम्नलिखित व्यंजक को सरल कीजिए:

$$\left[\frac{1}{8} + \left\{\frac{1}{6} \times \left(\frac{36}{45} \times \frac{25}{24}\right) - \left(\frac{12}{21} \times \frac{14}{15} \times \frac{45}{24}\right)\right\} + \frac{27}{36}\right]$$

SSC CGL Tier-II 26/10/2023

(a) $\frac{1}{27}$ **(b)** $\frac{1}{108}$

(c) $\frac{1}{72}$ **(d)** $\frac{1}{36}$

$\frac{1}{8} + \frac{1}{6} \times \frac{5}{6} - 1 + \frac{27}{36}$

$$-\overline{8+3} \qquad \overline{-8+3}$$

$$= -11 \qquad = -5$$

8. Simplify the following expression

दिए गए व्यंजक सरल कीजिए।

$$25-\left[16-\left\{14-\left(18-\frac{11}{8+3}\right)\right\}\right]$$

(a) 16 **(b) 18**

(c) 15 **(d) 20**

$25-[16-\{14-7\}]$

$=25-[16-7]$

$=25-9$

$=16$

$$3\frac{1}{3} = \frac{10}{3}$$

$$7\frac{+1}{\times 5} = \frac{36}{5}$$

$\frac{10}{3}-\left\{\frac{13}{3}+\left(\frac{10}{3}\div\overline{\frac{7}{3}-\frac{1}{3}}\right)\right\}$

$=\frac{10}{3}-\left\{\frac{13}{3}+\left(\frac{\cancel{10}^{5}}{\cancel{3}}\times\frac{\cancel{2}}{\cancel{6}_{3}}\right)\right\}$

$=\frac{10}{3}-\left\{\frac{13}{3}+\frac{5}{3}\right\}$

$=\frac{10}{3}-\frac{18}{3}=-\frac{8}{3}$

9. Simplify:

निम्न का मान ज्ञात कीजिए।

$$3\frac{1}{3}-\left\{4\frac{1}{3}+\left(3\frac{1}{3}\div\overline{2\frac{1}{3}-\frac{1}{3}}\right)\right\}$$

SSC CPO 04/10/2023 (Shift-01)

(a) $-\frac{8}{3}$

(b) $-\frac{1}{3}$

(c) $\frac{2}{3}$

(d) $\frac{1}{3}$

$A = \frac{9}{16} \times \frac{36}{25} 9 = \frac{81}{100}$ ✓

$B = 9 \times \frac{225}{4}$

$C = \frac{25}{64}$ $D = \frac{81}{100}$ ✓

$\frac{9}{16} \div \frac{25}{36}$

$\frac{9}{16} \times \frac{36}{25} 9$

10. If $A = \frac{9}{16} \div \frac{25}{36}$, $B = 9 \div [(16 \div 25) \div 36]$, $C = [9 \div (16 \div 25)] \div 36$, $D = (9 \div 16) \div (25 \div 36)$, then which of the following is true?

यदि $A = \frac{9}{16} \div \frac{25}{36}$, $B = 9 \div [(16 \div 25) \div 36]$, $C = [9 \div (16 \div 25)] \div 36$, $D = (9 \div 16) \div (25 \div 36)$ तो निम्नलिखित में से कौन-सा कथन सत्य है?

SSC CHSL 10/07/2024 (Shift-02)

(a) B and D are equal.

(b) C and D are equal.

(c) A and B are equal.

(d) A and D are equal.

CONCEPT-02

(A)

$$a^2 + b^2 = (a + b)^2 - 2ab$$

$$a^2 + b^2 = (a - b)^2 + 2ab$$

$$a^2 - b^2 = (a + b)(a - b)$$

* $(a+b)^2 = a^2 + b^2 + 2ab$

$$\therefore (a+b)^2 - 2ab = a^2 + b^2$$

* $(a-b)^2 = a^2 + b^2 - 2ab$

$$(a \cdot b)^2 + 2ab = a^2 + b^2$$

$$(a^2 - b^2) = (a+b)(a-b)$$

11. Simplify $52 \times 52 - 48 \times 48$

सरल कीजिए: $52 \times 52 - 48 \times 48$

SSC Phase-XI 27/06/2023 (Shift-03)

(a) 400 **(b) 386**

(c) 424 **(d) 326**

$a^2 - b^2 = (a+b)(a-b)$

$= 100 \times 4$

$= 400$

$= (1801)^2$

$= (1800+1)^2$

$= (1800)^2+(1)^2+2\times1800\times1$

$= 3240000+1+3600$

$= 3243601$

12. The value of 1801 ×1801 is:

1801 ×1801 का मान ज्ञात करें।

SSC CPO 09/11/2022 (Shift-02)

(a) 3423601 **(b) 3243601**

(c) 2343601 **(d) 3243106**

* $(101)^2 = (100+1)^2$

$= 10000+1+200$

$= 10201$

* $(1001)^2 = (1000+1)^2$

$= 1000000+1+2000$

$= 1002001$

* $(101)^2 = 10201$

* $(1001)^2 = 1002001$

* $(10001)^2 = 100020001$

1

13. Find the value of (1001+12)(1001–12).

(1001+12)(1001–12) का मान ज्ञात कीजिए

RRB Technician G-III, 29/12/2024 (Shift-02)

(a) 1000019857
(b) 1001857
(c) 1000190857
(d) 100019857

$(a+b)(a-b) = a^2 - b^2$

$(1001)^2 - (12)^2$
$= 1002001 - 144$
$= 1001857$

14. Simplify the following

निम्नलिखित को सरल करें।

$$\frac{4.35 \times 4.35 - 2.67 \times 2.67}{1.68}$$

RRB Technician G-III, 26/12/2024 (Shift-03)

(a) 7.02 (b) 6.99

(c) 7.05 (d) 6.96

$$\frac{a^2 - b^2}{1.68} = \frac{7.02 \times 1.68}{1.68}$$

$A = \frac{33 \times 83}{39 \times 83} = \frac{11}{13}$

$B = \frac{11 \times 41}{41 \times 71} = \frac{11}{71}$

$\therefore \frac{1}{B} - \frac{20}{A}$

$= \frac{71}{11} - \frac{13 \times 20}{11}$

$= \frac{11}{11} = 1$

15. If $A = \frac{58^2 - 25^2}{46^2 - 37^2}$, $B = \frac{26^2 - 15^2}{56^2 - 15^2}$, then the value of $\frac{1}{B} - \frac{20}{A}$ is:

यदि $A = \frac{58^2 - 25^2}{46^2 - 37^2}$, $B = \frac{26^2 - 15^2}{56^2 - 15^2}$, तो $\frac{1}{B} - \frac{20}{A}$ का मान है:

SSC CGL Tier-II 26/10/2023

(a) 2
(b) 1
(c) 0
(d) – 1

* $(a^2-b^2) = (a+b)(a-b)$

* $(a^4-b^4) = (a^2+b^2)(a+b)(a-b)$

* $(a^8-b^8) = (a^4+b^4)(a^2+b^2)(a+b)(a-b)$

* $(a^{16}-b^{16}) = (a^8+b^8)(a^4+b^4)(a^2+b^2)(a+b)(a-b)$

$(a^4-b^4) = (a^2)^2-(b^2)^2$

$= (a^2+b^2)(a^2-b^2)$

$= (a^2+b^2)(a+b)(a-b)$

16. Simplify the following expression:

निम्नलिखित व्यंजक को सरल कीजिए:

$$\frac{\left((32)^4 - 1\right)^4}{33 \times 31 \times \left(2^{10} + 1\right)}$$

SSC Phase-XII 26/06/2024 (Shift-04)

(a) 2 (b) 1

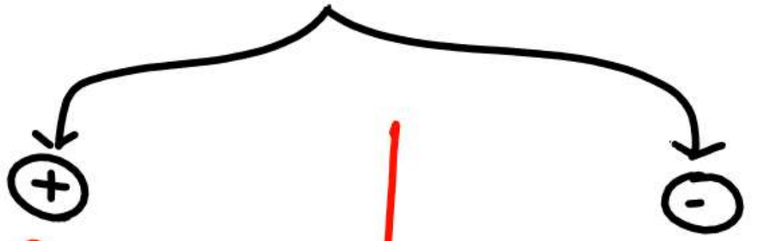
(c) –2 (d) –1

$$= \frac{[(32)^2+(1)^2][(32)^1+(1)^1][32-1]}{33 \times 31 \times (2^{10}+1)}$$

$$= \frac{1025}{1025} = 1$$

$$(a+b)^2 = a^2+b^2+2ab$$

$$(a-b)^2 = a^2+b^2-2ab$$



$(a+b)^2+(a-b)^2 = 2(a^2+b^2)$

$(a+b)^2-(a-b)^2 = 2\times 2ab = 4ab$

17. Simpify:

निम्न को सरल कीजिए।

$$\frac{(379+276)^2+(379-276)^2}{379\times379+276\times276}$$

$a+b \qquad a-b$

SSC CHSL 11/08/2023 (Shift-2)

(a) 2
(b) 655
(c) 103
(d) 1

$$\frac{(a+b)^2+(a-b)^2}{a^2+b^2}=\frac{2(a^2+b^2)}{a^2+b^2}$$

18. Simplify the following expression:

नम्नलिखित व्यंजक को सरल कीजिए:

$$\frac{(525 + 175)^2 - (525 - 175)^2}{(525 \times 175)}$$

SSC CPO 27/06/2024 (Shift-03)

(a) 350 **(b) 700**

(c) 2 **(d) 4**

$$\frac{(a+b)^2 - (a-b)^2}{ab} = \frac{2 \times 2ab}{ab}$$

19. Simplify: $\frac{(25)^4 + 1250 + 1 - (25)^2}{(25)^2 + 25 + 1}$

Hold

$\frac{(25)^4 + 1250 + 1 - (25)^2}{(25)^2 + 25 + 1}$ को सरल कीजिए।

SSC CGL 19/09/2024 (Shift-03)

(a) 602
(b) 601
(c) 603
(d) 604

20. The value of/का मान ज्ञात करें।

$$\frac{0.325\times0.325+0.175\times0.175+25\times0.00455}{5\times0.0065\times3.25-7\times0.175\times0.025}+\frac{0.5}{1.5}$$

SSC CPO 24/11/2020 (Shift-1)

(a) $\frac{11}{3}$ (b) 3

(c) 0 (d) $\frac{7}{3}$

$$\frac{a^2+b^2+2ab}{a^2-b^2}+\frac{1}{3}$$

$$=\frac{(a+b)^2}{(a-b)(a+b)}+\frac{1}{3}$$

$$=\frac{0.500}{0.150}+\frac{1}{3}=\frac{11}{3}$$

150

$\frac{a^2-b^2}{a^2+b^2+2ab}$

$= \frac{(a-b)(a+b)}{(a+b)(a+b)}$

0.334

2.338

7

21. The value of

$$\frac{4.669 \times 4.669 - 9 \times (0.777)^2}{(4.669)^2 + (2.331)^2 + 14(0.667)(2.331)}$$

is (1 – k), where k = ?

$$\frac{4.669 \times 4.669 - 9 \times (0.777)^2}{(4.669)^2 + (2.331)^2 + 14(0.667)(2.331)}$$ का मान (1 – k) है, जिसमें **k = ?**

0.334 = 1 - K

⇒ K = 0.666

SSC CPO 11/12/2019 (Shift-02)

(a) 0.666 ✓
(b) 0.647
(c) 0.467
(d) 0.768

$3^2 \times (0.777)^2$

$= (3 \times 0.777)^2$

$= (2.331)^2$

22. The value of $\dfrac{(0.13)^2+(0.21)^2}{(0.39)^2+81(0.07)^2} \div \dfrac{(2.4)^4+3\times(11.52)+9}{(2.4)^6+6(2.4)^4+3\times(17.28)}$ lies between:

का मान किसके बीच स्थित है?

SSC CPO 12/12/2019 (Shift-01)

(a) 0.7 and 0.8 (b) 0.4 and 0.5

(c) 0.6 and 0.7 (d) 0.5 and 0.6

$$\frac{a^2+b^2}{(3a)^2+(3b)^2} \div \frac{c^4+2c^2d+d^2}{c^6+2c^4d+c^2d^2}$$

$$= \frac{a^2+b^2}{9[a^2+b^2]} \div \frac{c^4+2c^2d+d^2}{c^2(c^4+2c^2d+d^2)}$$

$$= \frac{1}{9} \div \frac{1}{c^2} = \frac{1}{9} \times c^2 = \frac{2.4 \times 2.4}{9} = 0.64$$

22. The value of $\frac{(0.13)^2 + (0.21)^2}{(0.39)^2 + 81(0.07)^2} \div \frac{(2.4)^4 + 3 \times (11.52) + 9(3)^2}{(2.4)^6 + 6(2.4)^4 + 3 \times (17.28)}$ lies between:

का मान किसके बीच स्थित है?

SSC CPO 12/12/2019 (Shift-01)

(a) 0.7 and 0.8 (b) 0.4 and 0.5

(c) 0.6 and 0.7 (d) 0.5 and 0.6

$$\frac{a^2+b^2}{9(a^2+b^2)} \div \frac{c^4+2dc^2+d^2}{c^2(c^4+2dc^2+d^2)}$$

$$= \frac{1}{9} \div \frac{1}{c^2}$$

$$= \frac{1}{9} \times c^2 = \frac{0.8 \times 0.8}{9} = 0.64$$

(B)

$$a^3 + b^3 = (a + b)(a^2 - ab + b^2)$$

$$a^3 - b^3 = (a - b)(a^2 + ab + b^2)$$

* $(a^3+b^3) = (a+b)(a^2+b^2-ab)$

* $(a^3-b^3) = (a-b)(a^2+b^2+ab)$

23. The value of / का मान है:

$$\frac{(275\times275\times275+725\times725\times725)}{(275\times275+725\times725-725\times275)}$$

SSC CPO 29/06/2024 (Shift-02)

(a) –450000 (b) 45000

(c) 450000 (d) –45000

$$\left[\frac{a^3+b^3}{a^2+b^2-ab}\right]\times\left[\frac{a^3-b^3}{a^2+b^2+ab}\right]$$

$$=\frac{(a+b)(a^2+b^2-ab)}{(a^2+b^2-ab)}\times\frac{(a-b)(a^2+b^2+ab)}{(a^2+b^2+ab)}$$

$$=(a+b)(a-b)$$

$$=1000\times(-450)$$

$$=-450000$$

$$\frac{a^3-b^3}{a^2+ab+b^2}$$

$$= \frac{(a-b)(a^2+b^2+ab)}{(a^2+b^2+ab)}$$

$$= 0.73$$

24. Simplify: $\dfrac{(0.83)^3-(0.1)^3}{(0.83)^2+0.083+0.01} = ?$

समीकरण $\dfrac{\overset{a}{(0.83)}^3-\overset{b}{(0.1)}^3}{(0.83)^2+0.083+\underset{(0.1)^2}{0.01}} = ?$

SSC CHSL 14/08/2023 (Shift-2)

(a) 0.98 (b) 0.27

(c) 0.93 (d) 0.73

$$\begin{array}{r} 0.83 \\ -0.10 \\ \hline 0.73 \end{array}$$

25. Simplify the given expression:

दी गई अभिव्यक्ति को सरल कीजिए:

$$\frac{12.43\times12.43\times12.43+19.57\times19.57\times19.57}{12.43\times12.43-12.43\times19.57+19.57\times19.57}$$

SSC Phase-XII 24/06/2024 (Shift-03)

(a) 34 (b) 30

(c) 32 (d) 36

$$\frac{a^3+b^3}{a^2-ab+b^2}=\frac{(a+b)(a^2-ab+b^2)}{(a^2-ab+b^2)}=32$$

$$\frac{a^3+b^3}{\frac{a^2}{100}+\frac{b^2}{100}-\frac{ab}{100}}$$

$$= \frac{100\left[(a+b)(a^2+b^2-ab)\right]}{(a^2+b^2-ab)}$$

$$= 100 \times 1000$$

$$= 100000$$

26. $\dfrac{675\times675\times675+325\times325\times325}{67.5\times67.5+32.5\times32.5-67.5\times32.5}$ **is equal to:**

$$\frac{\overset{a}{675}\times675\times675+\overset{b}{325}\times325\times325}{\frac{67.5}{10}\times\frac{67.5}{10}+\frac{32.5}{10}\times\frac{32.5}{10}-\frac{67.5}{10}\times\frac{32.5}{10}}$$

निम्नलिखित में से किसके बराबर है:

(a) 100 (b) 10,000

(c) 1,000 (d) 1,00,000

26. $\dfrac{675\times675\times675+325\times325\times325}{67.5\times67.5+32.5\times32.5-67.5\times32.5}$ is equal to:

100

$\dfrac{a^3+b^3}{a^2+b^2-ab}$

निम्नलिखित में से किसके बराबर है:

(a) 100 (b) 10,000

(c) 1,000 (d) 1,00,000

$= 100 \times \dfrac{(a+b)(a^2+b^2-ab)}{(a^2+b^2-ab)}$

$= 100000$

27. The value of $\frac{(251)^3+(249)^3}{25.1\times25.1-624.99+24.9\times24.9}$ is 5×10^k, where the value of k is ____.

a^3 b^3

100 $\frac{(251)^3+(249)^3}{25.1\times25.1-624.99+24.9\times24.9}$ का मान 5×10^k, है, जहाँ **k** का मान ____ है।

a^2 ab b^2

(a) 4 (b) 5

(c) 3 (d) 6

$\Rightarrow 100\times\frac{(a+b)(a^2+b^2-ab)}{(a^2+b^2-ab)}=5\times10^k$

$\Rightarrow 50000=5\times10^k$

$\Rightarrow 5\times10^4=5\times10^k$

(C)

If $a + b + c = 0$

then $\Rightarrow a^3 + b^3 + c^3 = 3abc$

$a+b+c=0$

$a^3+b^3+c^3=3abc$

Q. $(3)^3 + (2)^3 - (5)^3 = ?$

(a) 30
(b) -30
(c) 0
(d) -90

m-1

$27 + 8 - 125$
$= 35 - 125$
$= -90$

Ans $(3)^3 + (2)^3 + (-5)^3$

Verify $3 + 2 - 5 = 0$

$= 3 \times 3 \times 2 \times (-5)$
$= -90$

$$= -\frac{0.74 \times 1.23 \times 0.13}{3 \times 0.37 \times 0.41 \times (-0.78)}$$

$$= -\frac{1}{3}$$

28. What is the value of/का मान ज्ञात कीजिए।

$$\frac{0.74 \times 1.23 \times 0.13}{(0.37)^3 + (0.41)^3 - 8(0.39)^3}$$

$+(-2)^3(0.39)^3$

SSC CPO 11/12/2019 (Shift-01)

$+(-0.78)^3$

(a) $\frac{-1}{3}$ (b) 1

(c) −1 (d) $\frac{1}{3}$

$0.37 + 0.41 - 0.78 = 0$

29. Simplify/सरल करें:

$$\frac{(3.321)^3 + (2.681)^3 + (1.245)^3}{(3.321)^2 + (2.681)^2 + (1.245)^2}$$

$$\frac{-3 \times 3.321 \times 2.681 \times 1.245}{-(3.321 \times 2.681) - (2.681 \times 1.245) - (1.245 \times 3.321)}$$

SSC CHSL 04/08/2023 (Shift-01)

(a) 6.125
(b) 8.645
(c) 7.247
(d) 10.245

$a^3+b^3+c^3-3abc$

$(a+b+c)(a^2+b^2+c^2-ab-bc-ac)$

$(a+b+c)\left[(a+b+c)^2-3(ab+bc+ac)\right]$

$\frac{(a+b+c)}{2}\left[3(a^2+b^2+c^2)-(a+b+c)^2\right]$

$\frac{(a+b+c)}{2}\left[(a-b)^2+(b-c)^2+(c-a)^2\right]$

$$\frac{(3.321)^3 + (2.681)^3 + (1.245)^3 - 3 \times 3.321 \times 2.681 \times 1.245}{(3.321)^2 + (2.681)^2 + (1.245)^2 - (3.321 \times 2.681) - (2.681 \times 1.245) - (1.245 \times 3.321)}$$

SSC CHSL 04/08/2023 (Shift-01)

(a) 6.125 (b) 8.645
(c) 7.247 (d) 10.245

$$\frac{a^3+b^3+c^3-3abc}{a^2+b^2+c^2-ab-bc-ac} = \frac{(a+b+c)(a^2+b^2+c^2-ab-bc-ac)}{(a^2+b^2+c^2-ab-bc-ac)}$$

$$= 7.247$$

30. Simplify the given expression.

$$\frac{(80\times80\times80)+(70\times70\times70)+(50\times50\times50)-840000}{6400+4900+2500-5600-3500-4000}$$

दिए गए व्यंजक को सरल कीजिए।

SSC CHSL 10/08/2023 (Shift-01)

(a) 100 (b) 200
(c) 400 (d) 300

a^3 b^3 c^3

$$\frac{a^3+b^3+c^3-3abc}{a^2+b^2+c^2-ab-bc-ac} = (a+b+c) = 200$$

31. Simplify the following.

निम्नलिखित का सरलीकरण कीजिए।

$$\frac{0.01\times0.01\times0.01+0.003\times0.003\times0.003}{0.05\times0.05-0.015\times0.05+0.015\times0.015}$$

SSC CGL 24/07/2023 (Shift-3)

(a) $\frac{13}{25}\times10^{3}$

(b) $\frac{13}{15}\times10^{-3}$

(c) $\frac{13}{15}\times10^{3}$

(d) $\frac{13}{25}\times10^{-3}$

$a^3+b^3 = (a+b)(a^2-ab+b^2)$

$5\times5(0.01\times0.01-0.003\times0.01+0.003\times0.003)$

$25(a^2-ab+b^2)$

$= \frac{(a+b)(a^2+b^2-ab)}{25(a^2+b^2-ab)}$

$= \frac{0.013}{25}\times\frac{1}{1000}$

$= \frac{13}{25}\times10^{-3}$

$0.01 + 0.003 = 0.013$

Bar वाले सवाल

1) $0.\overline{7}$, $0.\overline{8}$, $0.\overline{5}$, $0.\overline{63}$, $0.1\overline{23}$, $0.\overline{3675}$

2) $0.9\overline{7}$, $0.1\overline{5}$, $0.1\overline{23}$, $0.12\overline{3}$

3) $1.\overline{23}$, $1.2\overline{3}$

4) $\overline{1}.23$, $\overline{2}.57$

$*$ $x = 0.55555\ldots\ldots\infty$

Ans $x = 0.55555\ldots\ldots\infty$ —— (i)

$10x = 5.5555\ldots\infty$ —— (ii)

$\Rightarrow 9x = 5.0$

$\Rightarrow x = \frac{5}{9}$

Type-1

$x = 0.\bar{5}$ $\quad$ $x = 0.\overline{15}$ $\quad$ $x = 0.\overline{23}$ $\quad$ $x = 0.\overline{67}$

$\frac{5}{9}$ $\quad$ $\frac{15}{99}$ $\quad$ $\frac{23}{99}$ $\quad$ $\frac{67}{99}$

Type-2

$x = 0.1\bar{2}$ $\quad$ $0.7\bar{3}$ $\quad$ $x = 0.6\bar{7}$ $\quad$ $x = 0.5\bar{9}$

$\frac{12-1}{90} = \frac{11}{90}$ $\quad$ $\frac{73-7}{90} = \frac{66}{90}$ $\quad$ $\frac{67-6}{90} = \frac{61}{90}$ $\quad$ $\frac{59-5}{90} = \frac{54}{90}$

$0.\overline{123}$ $0.1\overline{23}$ $0.12\overline{3}$

ans → $\frac{123}{999}$ $\frac{123-1}{990} = \frac{122}{990}$ $= \frac{123-12}{900} = \frac{111}{900}$

Type-3 $1.\overline{23}$ $1.2\overline{3}$ $12.\overline{34}$ $12.3\overline{4}$

ans $\frac{123-1}{99}$

$= \frac{122}{99}$

$= 1 + 0.2\overline{3}$

$= 1 + \frac{23-2}{90}$

$= 1 + \frac{21}{90}$

$= \frac{111}{90}$

$\frac{123-12}{90}$

$= \frac{111}{90}$

$= \frac{1234-12}{99}$

$= \frac{1222}{99}$

$= \frac{1234-123}{90}$

$= \frac{1111}{90}$

CONCEPT-03 (Bar Type Questions)

$= 0.\overline{7}$

$= \frac{7}{9}$

32. Convert 0.7777..... ∞ into fraction

दशमलव 0.7777..... ∞ को भिन्न में बदलें

(a) $\frac{7}{9}$

(b) $\frac{7}{3}$

(c) $\frac{7}{10}$

(d) $\frac{77}{99}$

33. Convert 0.535353.....∞ into fraction

दशमलव 0.535353.....∞ को भिन्न में बदलें

(a) $\frac{53}{99}$ **(b)** $\frac{53}{49}$

(c) $\frac{53}{100}$ **(d)** $\frac{53}{59}$

$= 0.\overline{53}$

$= \frac{53}{99}$

34. Convert it into vulgar fraction

$0.5\overline{87}$

$$\frac{587-5}{990} = \frac{582}{990} \quad (97,\ 165)$$

$$= \frac{97}{165}$$

CISF HCM 30/10/2023 (Shift-01)

(a) $\frac{93}{167}$

(b) $\frac{97}{165}$

(c) $\frac{95}{167}$

(d) $\frac{91}{165}$

35. Simplify $8.\overline{46}$ in the fractional form.

$8.\overline{46}$ को एक अभ्यस्त भिन्न के रूप में सरल बनाएं

$8.\overline{46}$

$= \frac{846 - 8}{99} = \frac{838}{99}$

RRB Technician G-III, 26/12/2024 (Shift-01)

(a) $\frac{838}{99}$

(b) $\frac{83}{99}$

(c) $\frac{846}{99}$

(d) $\frac{84}{99}$

$0.0\overline{654}$

$= \frac{654-6}{9900} = \frac{648}{9900}$

18
~~108~~
~~1650~~
275

$= \frac{18}{275}$

36. Correct expression of $0.0\overline{654}$. (the bar indicates repeating decimal)

$0.0\overline{654}$ का सही व्यंजक है (बार दशमलव की पुनरावृतित्त को दर्शाता है):

NTPC CBT-2 19/01/2017 (Shift-1)

(a) $\frac{654}{1000}$

(b) $\frac{654}{10000}$

(c) $\frac{18}{275}$ ✓

(d) $\frac{18}{277}$

37. Correct expression of $1.4\overline{27}$. (the bar indicates repeating decimal)

$1.4\overline{27}$ का सही सरलीकरण है (बार दशमलव की पुनरावृत्ति को दर्शाता है):

$$\frac{1427-14}{990}$$

$$= \frac{1413}{990} \quad \frac{157}{110}$$

NTPC CBT-2 17/01/2017 (Shift-3)

(a) $\frac{1427}{1000}$

(b) $\frac{157}{110}$

(c) $\frac{1427}{10000}$

(d) $\frac{157}{111}$

38. $2.\overline{8768} = ?$

$$\frac{28768-28}{9990}$$

$$= \frac{28740}{9990}$$

$$= 2\frac{876}{999}\ \frac{292}{333}$$

$$= 2\frac{292}{333}$$

(a) $2\frac{878}{999}$

(b) $2\frac{9}{10}$

(c) $2\frac{292}{333}$

(d) $2\frac{4394}{4995}$

39. Find the Value of x/x का मान ज्ञात कीजिये

$$0.\bar{3} + 0.\bar{4} + 0.\bar{5} + 0.\bar{6} = x$$

(a) 3 (b) 5

(c) 2 (d) 8

$$\frac{3}{9} + \frac{4}{9} + \frac{5}{9} + \frac{6}{9}$$

$$= \frac{18}{9} = 2$$

40. The value of $0.\overline{56} + 0.\overline{43} + 0.\overline{89}$ is

$0.\overline{56} + 0.\overline{43} + 0.\overline{89}$ का मान है

NTPC CBT-1, 23/02/2021 (Shift-01)

(a) $1.\overline{98}$ $= \frac{198-1}{99} = \frac{197}{99}$

(b) $1.\overline{87}$

(c) $1.\overline{89}$ $\frac{188}{99}$

(d) $1.\overline{88} = \frac{188-1}{99}$

$= \frac{187}{99}$

$\frac{56}{99} + \frac{43}{99} + \frac{89}{99}$

$= \frac{188}{99}$

40. The value of $0.\overline{56} + 0.\overline{43} + 0.\overline{89}$ is

$0.\overline{56} + 0.\overline{43} + 0.\overline{89}$ का मान है

NTPC CBT-1, 23/02/2021 (Shift-01)

(a) $1.\overline{98}$

(b) $1.\overline{87}$

(c) $1.\overline{89}$

(d) $1.\overline{88}$

0.565656
0.434343
0.898989
1.898988

Ans = $1.\overline{89}$

41. $3.2\overline{45} + 1.2\overline{34} - 2.\overline{12}$ **is equal to:**

$3.2\overline{45} + 1.2\overline{34} - 2.\overline{12}$ बराबर है -

ICAR Mains, 08/07/2023 (Shift-3)

(a) $2.3\overline{58}$ $\frac{2358-23}{990}$ $\frac{2335}{990}$ ✓

(b) $2.4\overline{37}$

(c) $2.2\overline{43}$

(d) $2.5\overline{36}$

$$= \frac{3245-32}{990} + \frac{1234-12}{990} - \frac{(212-2)}{99}$$

$$= \frac{3213}{990} + \frac{1222}{990} - \frac{2100}{990}$$

$$= \frac{2335}{990}$$

41. $3.2\overline{45} + 1.2\overline{34} - 2.\overline{12}$ is equal to:

$3.2\overline{45} + 1.2\overline{34} - 2.\overline{12}$ बराबर है -

ICAR Mains, 08/07/2023 (Shift-3)

(a) $2.3\overline{58}$ (b) $2.4\overline{37}$

(c) $2.2\overline{43}$ (d) $2.5\overline{36}$

$3.2454545\overline{4}$

$+\ 1.2343434\overline{3}$

$-\ 2.1212121\overline{2}$

$2.3585858\overline{5}$

$= 2.3\overline{58}$

$$\frac{123-12}{900} + \frac{39}{99}$$

$$= \frac{111}{900} + \frac{39}{99}$$

$$= \frac{1221+3900}{9900} = \frac{5121}{9900} = \frac{569}{1100}$$

42. Express the following into vulgar fraction.

निम्नलिखित को साधारण भिन्न में व्यक्त करें।

$0.\overline{123} + 0.\overline{39}$

RPF SI 12/12/2024 (Shift-01)

(a) $\frac{568}{1100}$

(b) $\frac{569}{1100}$

(c) $\frac{578}{1100}$

(d) $\frac{579}{1100}$

$$\frac{215}{90} \div \frac{49}{90}$$

$$= \frac{215}{\cancel{90}} \times \frac{\cancel{90}}{49}$$

$$= 4\frac{19}{49}$$

43. Find the value of $2.3\overline{8} \div 0.5\overline{4}$.

निम्नांकित का मान ज्ञात कीजिए।

RRB JE 18/12/2024 (Shift-02)

(a) $4\frac{19}{49}$

(b) $3\frac{17}{21}$

(c) $5\frac{26}{51}$

(d) $6\frac{29}{57}$

$$\begin{array}{r} 238 \\ -23 \\ \hline 215 \end{array}$$

44. Find the value of/का मान ज्ञात कीजिये

$$0.\overline{2} + 0.\overline{3} + 0.\overline{32}$$

(a) $0.\overline{77}$ (b) $0.\overline{82}$

(c) $0.\overline{86}$ (d) $0.\overline{87}$

0.222222
0.333333
0.323232

0.878787

$= 0.\overline{87}$

$\left(\frac{324786-324}{999}\right)-\left(\frac{10193-1019}{900}\right)$

45. Find the value of/का मान ज्ञात कीजिये

$$324.\overline{786}-10.19\overline{3}$$

(a) $314.59\overline{345}$ (b) $314.59\overline{445}$

(c) $314.59\overline{345}$ (d) $314.59\overline{445}$

$$= \frac{324462}{999} - \frac{9174}{900}$$

$$= \frac{10815400-339438}{33300} = \frac{10475962}{33300}$$

$$= \frac{31427886}{99900}$$

$$= \frac{31427886}{99900}$$

$$\begin{array}{r} 314593457 \\ -\ 31459 \\ \hline 31427886 \end{array}$$

45. Find the value of/का मान ज्ञात कीजिये

$$324.\overline{786} - 10.19\overline{3}$$

(a) $314.59\overline{345}$ (b) $314.59\overline{445}$

(c) $314.59\overline{345}$ (d) $314.59\overline{445}$

$$\begin{array}{r} 0.786786786 \\ -\ 0.193333333 \\ \hline 0.593453453 \end{array}$$

$$= 0.59\overline{345}$$

46. If $A = 0.3\overline{12}$, $B = 0.4\overline{15}$ and $C = 0.30\overline{9}$, then what is the value of A + B + C?

यदि $A = 0.3\overline{12}$, $B = 0.4\overline{15}$ तथा $C = 0.30\overline{9}$ है, तो $A + B + C$ का मान कितना है?

(a) $\frac{1211}{1100}$ (b) $\frac{1043}{1100}$

(c) $\frac{1097}{1100}$ (d) $\frac{1141}{1100}$

0.3121212
0.41515151
0.30999999
1.03727260

$= 1.03\overline{72}$

$\frac{10372 - 103}{9900} = \frac{10269}{9900}$ $\frac{1141}{1100}$

47. Find the value of/का मान ज्ञात कीजिये

$22.\overline{4} + 11.5\overline{67} - 33.5\overline{9}$

SSC CGL TIER - II 11/09/2019

$$\begin{array}{r} 22.44444444 \\ +\ 11.56767677 \\ -\ 33.59999999 \\ \hline 0.41212112 \end{array}$$

$= 0.4\overline{12}$

(a) $0.\overline{412}$ (b) $0.3\overline{1}$

(c) $0.4\overline{12}$ (d) $0.\overline{32}$

Bar → 2
No Bar → 1

ans = 0.$\overline{\quad}$

47. Find the value of/का मान ज्ञात कीजिये

$22.\overline{4} + 11.5\overline{67} - 33.5\overline{9}$

SSC CGL TIER - II 11/09/2019

(a) $0.\overline{412}$
(b) $0.3\overline{1}$
(c) $0.41\overline{2}$
(d) $0.\overline{32}$

2 points

) Max^m कितने Digits पर बार है?
) " " " " " नहीं है?

48. Find the value of/का मान ज्ञात कीजिये

$$0.\overline{57} - 0.4\overline{32} + 0.3\overline{5}$$

SSC CGL TIER - II 16/11/2020

(a) $0.\overline{494}$

(b) $0.\overline{498}$

(c) $0.\overline{498}$

(d) $0.\overline{494}$

$$\begin{array}{r} 0.57575 \\ +\ 0.35555 \\ -\ 0.43232 \\ \hline 0.49898 \end{array}$$

$$= 0.\overline{498}$$

49. Find the value of/का मान ज्ञात कीजिये

$$0.4\overline{7} + 0.5\overline{03} - 0.3\overline{9} \times 0.\overline{8}$$

SSC CGL TIER - II 13/09/2019

(a) $0.6\overline{15}$ (b) $0.\overline{615}$

(c) $0.62\overline{5}$ (d) $0.6\overline{25} = \frac{619}{990}$

$$\frac{43}{90} + \frac{498}{990} - \frac{36}{90} \times \frac{8}{9}$$

$$= \frac{11}{90} + \frac{498}{990}$$

$$= \frac{121 + 498}{990} = \frac{619}{990}$$

50. **Find the value of/का मान ज्ञात कीजिये**

HW

$$0.5\bar{6} - 0.7\overline{23} + 0.3\bar{9} \times 0.\bar{7}$$

SSC CGL TIER - II 12/09/2019

(a) $0.1\overline{54}$

(b) $0.\overline{154}$

(c) $0.\overline{158}$

(d) $0.1\overline{58}$

51. Find the value of/का मान ज्ञात कीजिये

$$\left(2.\overline{4} \times 0.\overline{6} \times 3 \times 0.1\overline{6}\right) \times \left[0.\overline{27} \times \left(0.8\overline{3} \div 0.1\overline{6}\right)\right]$$

SSC CGL TIER - II 15/11/2020

(a) $0.\overline{814}$ (b) $0.\overline{11}$

(c) $1.\overline{1} = \frac{10}{9}$ (d) $1.\overline{36}$

$$= \frac{22}{9} \times \frac{6}{9} \times 3 \times \frac{18}{90} \times \frac{21}{99} \times \frac{78}{90} \times \frac{90}{18}$$

$$= \frac{10}{9}$$

① $1.\overline{32}$

Soln $\frac{132-1}{99}$

$= \frac{131}{99}$

② $\bar{1}.32$

Soln $-1+0.32$

$= -0.68$

③ $\bar{2}.34$

Ans $-2+0.34$

$= -1.66$

$$\begin{array}{r} -1.00 \\ +\ 0.32 \\ \hline 0.68 \end{array}$$

52. $\bar{2}.75 + \bar{3}.78$

(a) $\bar{1}.03$ (b) $\bar{1}.53$

(c) $\bar{4}.53$ (d) $\bar{5}.53$

$-4+0.53$

$= -3.47$

$-2+0.75-3+0.78$

$= -5+1.53$

$= -3.47$

53. If $0.\overline{ab} + 0.\overline{ba} = \frac{5}{9}$, find the value of a + b.

यदि $0.\overline{ab} + 0.\overline{ba} = \frac{5}{9}$, तो a + b का मान ज्ञात कीजिये।

(a) 5 (b) 6
(c) 7 (d) 8

$$\frac{ab-a}{90} + \frac{ba-b}{90} = \frac{5}{9}$$

$$\Rightarrow \frac{10a+b-a+10b+a-b}{90} = \frac{5}{9}$$

$$\Rightarrow \frac{10(a+b)}{90} = \frac{5}{9}$$

$$xy = x \times 10 + y \times 1$$

$$xyz = x \times 100 + y \times 10 + z \times 1$$

$$ab = a \times 10 + b \times 1$$

$$= 10a + b$$

54. If $0.\overline{xy} = \frac{7}{11}$, find $x^2 + y^2 = ?$

यदि $0.\overline{xy} = \frac{7}{11}$, तब $x^2 + y^2 = ?$

(a) 36 (b) 44
(c) 45 (d) 55

$\frac{xy}{99} = \frac{7}{11}$

$\Rightarrow xy = 63$

$x^2+y^2 = 6^2+3^2$

$= 36+9$

$= 45$

Comparison of Fractions

1) Base Method / LCM Method
2) Difference Method
   - P
   - I
3) Cross Multiplication
4) Approximation

* Base/LCM Method *

Q. $\frac{2}{3}, \frac{3}{5}, \frac{11}{15}$

LCM (3, 5, 15) = 15

Ans

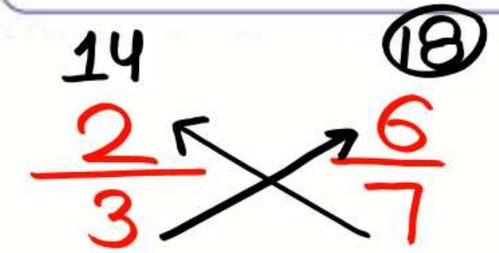
$\frac{2\times5}{3\times5}, \frac{3\times3}{5\times3}, \frac{11}{15}$

$\frac{10}{15}, \frac{9}{15}, \frac{11}{15}$

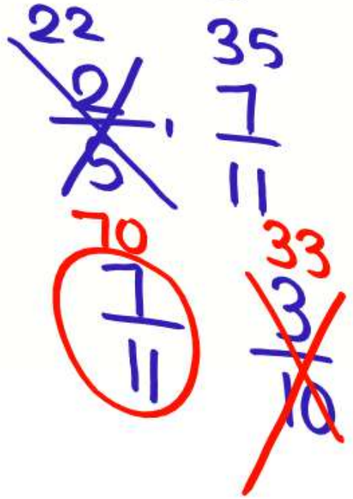
S

L

Cross Multiplication Method



$$\frac{2}{5} \cdot \frac{7}{11} , \frac{3}{10}$$



60 77
5 7
11 12

65 88
5 8
11 13

85 91
5 7
11 11

55. **Which fraction among the following is the least?**

निम्नलिखित में से कौन-सा भिन्न सबसे छोटा है?

$\frac{5}{11}, \frac{7}{12}, \frac{8}{13}, \frac{9}{17}$

SSC CGL MAINS (08/08/2022)

(a) $\frac{8}{13}$

(b) $\frac{5}{11}$

(c) $\frac{9}{17}$

(d) $\frac{7}{12}$

55. Which fraction among the following is the least?

निम्नलिखित में से कौन-सा भिन्न सबसे छोटा है?

85 99

$\frac{5}{11}, \frac{7}{12}, \frac{8}{13}, \frac{9}{17}$

SSC CGL MAINS (08/08/2022)

(a) $\frac{8}{13}$

(b) $\frac{5}{11}$

(c) $\frac{9}{17}$

(d) $\frac{7}{12}$

56. Find the greatest of the following fractions.

निम्नलिखित भिन्नों में से सबसे बड़ी भिन्न ज्ञात कीजिए।

$\frac{8}{11}, \frac{15}{19}, \frac{4}{5}, \frac{13}{21}$

CRPF HCM 23/02/2023 (Shift - 01

(a) $\frac{13}{21}$

(b) $\frac{15}{19}$

(c) $\frac{4}{5}$

(d) $\frac{8}{11}$

57. Which of the following is the largest fraction?

निम्न में से सबसे बड़ी भिन्न कौन सी है?

120 117

$\frac{8}{9}, \frac{6}{11}, \frac{4}{9}, \frac{13}{15}$

(a) $\frac{8}{9}$

(b) $\frac{6}{11}$

(c) $\frac{4}{9}$

(d) $\frac{13}{15}$

LCM (2,4,7) = 28

$\frac{1}{2}, \frac{5}{7}, \frac{3}{4}, \frac{6}{7}$

$\frac{14}{28}, \frac{20}{28}, \frac{21}{28}, \frac{24}{28}$ G

LCM METHOD

58. Find the greatest among

निम्नलिखित भिन्नों में से सबसे बड़ी भिन्न ज्ञात कीजिए।

$\frac{1}{2}, \frac{5}{7}, \frac{3}{4}$ & $\frac{6}{7}$

(a) $\frac{1}{2}$

(b) $\frac{5}{7}$

(c) $\frac{3}{4}$

(d) $\frac{6}{7}$ ✓

59. Find the smallest among

निम्नलिखित में से कौन-सा भिन्न सबसे छोटा है?

$\frac{18}{27}, \frac{24}{27}, \frac{10}{27}, \frac{48}{27}$

$\frac{2}{3}, \frac{8}{9}, \frac{10}{27}$ & $\frac{16}{9}$

(a) $\frac{2}{3}$

(b) $\frac{8}{9}$

(c) $\frac{10}{27}$

(d) $\frac{16}{9}$

Difference method

Fraction

Proper (N<D)

S: 2 — 4 — L: 6

$1\left(\frac{2}{3}\right)$ $1\left(\frac{4}{5}\right)$ $1\left(\frac{6}{7}\right)$

8: 4×3, 2×4, 5

9: 3×4, 3×3, 7

42: 6×2, 7×6, 9

Improper (N>D)

$\frac{3}{2}$, $\frac{5}{4}$, $\frac{7}{6}$

PROPER FRACTIONS

Numerator of the fraction is less then denominator or we can say value of the fraction is less than 1.

अंश का मान हर से छोटा हो अथवा भिन्न का मान 1 से कम हो।

$\frac{1}{2}, \frac{2}{3}, \frac{4}{5}, \frac{7}{11}, \frac{12}{23}$ **etc.**

To compare/तुलना के लिए:

Step 1: Take the difference of N^r and D^r of each of the fractions. /प्रत्येक भिन्न के अंश तथा हर का अंतर ज्ञात करें।

Step 2: Difference must be same. If the given difference is not same, make them same by taking LCM of each difference./अंतर समान होना चाहिए। यदि अंतर समान नही हैं तो प्रत्येक अंतर का ल.स.प. लेकर उसे समान कर लें।

Step 3: Fraction with smaller N^r will be least and fraction with greater N^r will be greatest./
इस प्रकार प्राप्त न्यूनतम अंश वाली भिन्न सबसे छोटी तथा अधिकतम अंश वाली भिन्न सबसे बड़ी होगी।

Ex: Compare/तुलना करें।

S L

$1\left(\frac{4}{5}, 1\left(\frac{6}{7}, 1\left(\frac{13}{14}, 1\left(\frac{11}{12}\right.\right.\right.\right.$

L S

$1\left(\frac{8997}{8998}\right.$ $1\left(\frac{6231}{6232}\right.$

Ex: The greatest value among the fractions

सबसे बड़ा और सबसे छोटा भिन्न ज्ञात करें।

$$\frac{2}{7}, \frac{1}{3}, \frac{5}{6}, \frac{3}{4}$$

L

4 5 50 30

2×5 $\frac{2 \times 2}{7}$ 5×2 $\frac{1 \times 5}{3}$ 1 $\frac{50}{6}$ 1 $\frac{30}{4}$

105 147 96 154

4 (5/9) 4 (7/11) 7 (8/15) 6 (11/17)

5 1

LCM(4,7,6) = 84

diff = 11/17 − 8/15

= (165 − 136)/255

= 29/255

60. What is difference between the largest and the smallest fractions among $\frac{5}{9}, \frac{7}{11}, \frac{8}{15}$ and $\frac{11}{17}$?

$\frac{5}{9}, \frac{7}{11}, \frac{8}{15}$ और $\frac{11}{17}$ में से सबसे बड़े और सबसे छोटे भिन्न का अंतर क्या है?

CRPF HCM 24/02/2023 (Shift - 02)

(a) $\frac{29}{255}$ ✓

(b) $\frac{8}{99}$

(c) $\frac{1}{45}$

(d) $\frac{17}{165}$

61. **What is the difference of the largest and smallest of the given fractions?**

H.W

दी गई भिन्नों में से सबसे बड़ी और सबसे छोटी भिन्न का अंतर क्या है?

$\frac{5}{11}, \frac{5}{7}, \frac{3}{8}, \frac{6}{11}$

SSC CHSL 13/03/2023 (Shift-01)

(a) $\frac{17}{56}$

(b) $\frac{19}{56}$

(c) $\frac{1}{7}$

(d) $\frac{23}{56}$

IMPROPER FRACTIONS

Numerator is greater than denominator or value of the fraction is greater than 1.

अंश का मान हर से बड़ा हो अथवा भिन्न का मान 1 से अधिक हो।

$\frac{3}{2}, \frac{13}{4}, \frac{6}{5}, \frac{27}{17}$ **etc.**

To compare/तुलना के लिए:

Step 1: Take the differecne of N^r and D^r of each of the fractions./प्रत्येक भिन्न के अंश तथा हर का अंतर ज्ञात करें।

Step 2: Difference must be same. If the given difference is not same, make them same by taking LCM of each difference./अंतर समान होना चाहिए। यदि अंतर समान नही हैं तो प्रत्येक अंतर का ल.स.प. लेकर उसे समान कर लें।

Step 3: Fraction with smaller N^r will be greatest and fraction with greater N^r will be smallest./इस प्रकार प्राप्त न्यूनतम अंश वाली भिन्न सबसे बड़ी तथा अधिकतम अंश वाली भिन्न सबसे छोटी होगी।

Proper Fraction

S

$1\frac{2}{3}$

L

$1\frac{7}{8}$

Improper Fraction

L

$1\frac{8}{7}$

S

$1\frac{11}{10}$

Ex: Compare/तुलना करें।

$$\frac{23}{18}, \frac{37}{32}$$

L S

$$5\left(\frac{23}{18}\right) > 5\left(\frac{37}{32}\right)$$

Ex: Find smallest and greatest fraction

सबसे छोटा और सबसे बड़ा भिन्न ज्ञात करें।

L S

$$\frac{16}{15}, \frac{20}{19}, \frac{25}{24}, \frac{35}{34}$$

Ex: Find smallest and greatest fraction

सबसे छोटा और सबसे बड़ा भिन्न ज्ञात करें।

$$\frac{15}{16}, \frac{8}{3}, \frac{11}{12}, \frac{7}{8}$$

L S

Proper < Improper

BASE METHOD

(i) When denominator is equal.

जब हर बराबर हो।

$$\frac{2}{17}, \frac{14}{17}, \frac{9}{17}, \frac{25}{17}$$

Fraction with greater numerator will be greatest and vice-versa.

बड़े अंश वाला भिन्न सबसे बड़ा होगा और इसके विपरीत भी सही होगा।

(ii) When Numerator is equal.

जब अंश बराबर हो।

$$\frac{9}{4}, \frac{9}{7}, \frac{9}{10}, \frac{9}{13}$$

Fraction with smaller denominator will be greatest and vice-versa.

छोटे हर वाला भिन्न सबसे बड़ा होगा और इसका विपरीत सही होगा।

(iii) If we increase N^r and decrease D^r, then resultant fraction will be greater.

यदि हम N^r को बढ़ाते हैं और D^r को घटाते हैं, तो परिणामी भिन्न अधिक होगा।

e.g. (i) $\frac{3}{7} < \frac{5}{6}$

(iii) If we increase N^r and decrease D^r, then resultant fraction will be greater.

यदि हम N^r को बढ़ाते हैं और D^r को घटाते हैं, तो परिणामी भिन्न अधिक होगा।

(ii) $\frac{101}{236} < \frac{103}{234}$

(iii) If we increase N^r and decrease D^r, then resultant fraction will be greater.

यदि हम N^r को बढ़ाते हैं और D^r को घटाते हैं, तो परिणामी भिन्न अधिक होगा।

(iii) $\frac{339}{237} < \frac{347}{231}$

(iv) If we decrease N^r and increase D^r then resultant fraction will be smaller.

यदि हम N^r को घटाएँ और D^r को बढ़ाएँ तो परिणामी भिन्न छोटा हो जाएगा।

e.g. (i) $\frac{7}{11} > \frac{6}{13}$

(iv) If we decrease N^r and increase D^r then resultant fraction will be smaller.

यदि हम N^r को घटाएँ और D^r को बढ़ाएँ तो परिणामी भिन्न छोटा हो जाएगा।

(ii) $\frac{101}{236} > \frac{99}{247}$

(iv) If we decrease N^r and increase D^r then resultant fraction will be smaller.

यदि हम N^r को घटाएँ और D^r को बढ़ाएँ तो परिणामी भिन्न छोटा हो जाएगा।

(iii) $\frac{334}{229} > \frac{329}{235}$

(v) **If we increase N^r and D^r together or we decrease N^r and D^r together than the resultant fraction can be increase or decrease or will have no change that can be determined by using percentage change.**

यदि हम N^r और D^r को एक साथ बढ़ाते हैं या हम N^r और D^r को एक साथ घटाते हैं तो परिणामी भिन्न बढ़ या घट सकता है या इसमें कोई परिवर्तन नहीं होगा जिसे प्रतिशत परिवर्तन का उपयोग करके निर्धारित किया जा सकता है।

e.g. (i) $\frac{123}{237}$ & $\frac{137}{267}$

(v) **If we increase N^r and D^r together or we decrease N^r and D^r together than the resultant fraction can be increase or decrease or will have no change that can be determined by using percentage change.**

यदि हम N^r और D^r को एक साथ बढ़ाते हैं या हम N^r और D^r को एक साथ घटाते हैं तो परिणामी भिन्न बढ़ या घट सकता है या इसमें कोई परिवर्तन नहीं होगा जिसे प्रतिशत परिवर्तन का उपयोग करके निर्धारित किया जा सकता है।

(ii) $\frac{423}{322}$ $\frac{492}{389}$

(v) **If we increase N^r and D^r together or we decrease N^r and D^r together than the resultant fraction can be increase or decrease or will have no change that can be determined by using percentage change.**

यदि हम N^r और D^r को एक साथ बढ़ाते हैं या हम N^r और D^r को एक साथ घटाते हैं तो परिणामी भिन्न बढ़ या घट सकता है या इसमें कोई परिवर्तन नहीं होगा जिसे प्रतिशत परिवर्तन का उपयोग करके निर्धारित किया जा सकता है।

(iii) $\frac{247}{437}$ $\frac{228}{387}$

(v) **If we increase N^r and D^r together or we decrease N^r and D^r together than the resultant fraction can be increase or decrease or will have no change that can be determined by using percentage change.**

यदि हम N^r और D^r को एक साथ बढ़ाते हैं या हम N^r और D^r को एक साथ घटाते हैं तो परिणामी भिन्न बढ़ या घट सकता है या इसमें कोई परिवर्तन नहीं होगा जिसे प्रतिशत परिवर्तन का उपयोग करके निर्धारित किया जा सकता है।

(iv) $\frac{743}{829}$ $\frac{691}{789}$

Approximation

(Base Method)

S L

$\frac{2}{7}, \frac{3}{7}, \frac{4}{7}, \frac{5}{7}$

$D^r \rightarrow$ Same

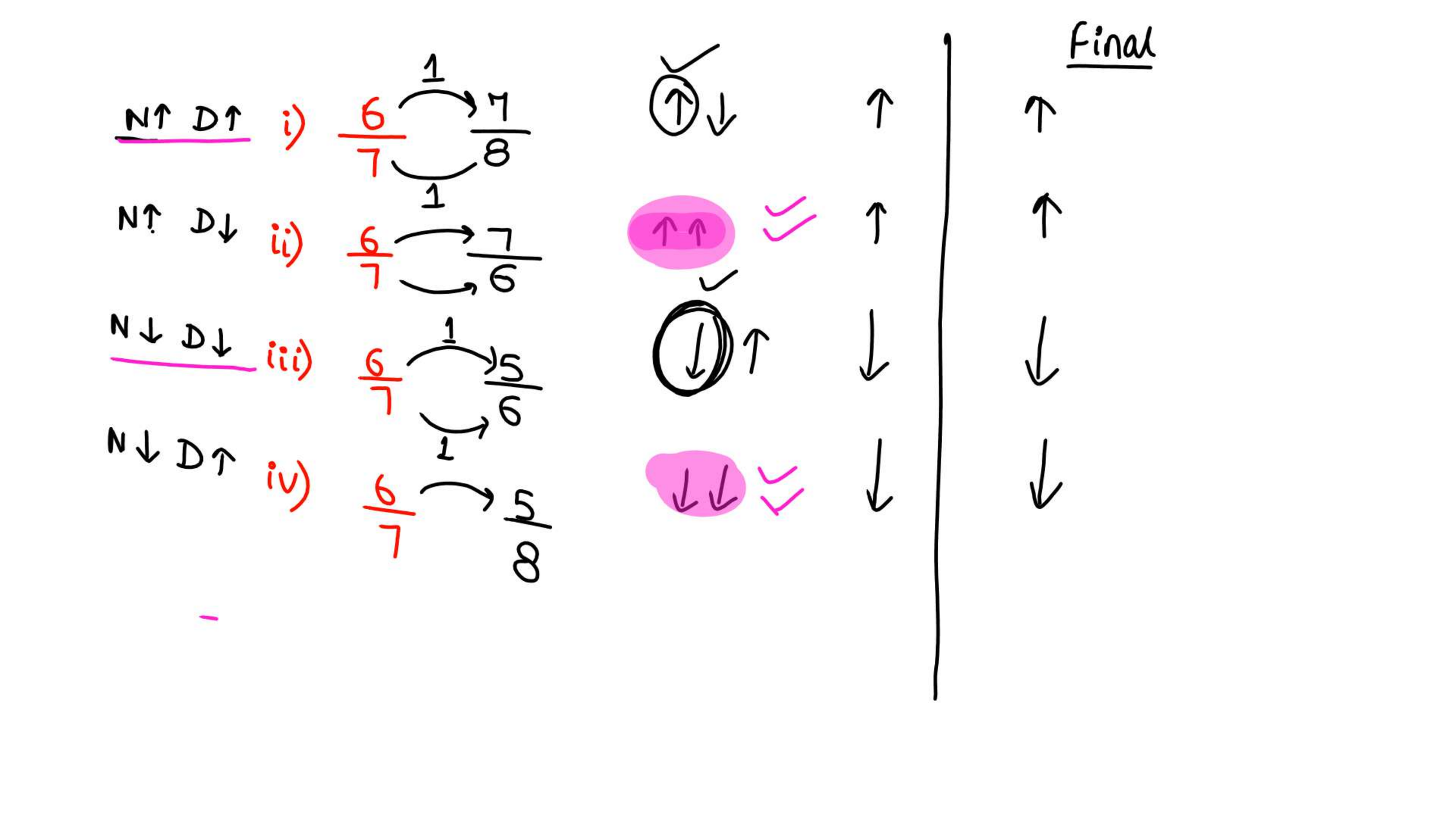
N↑ V↑

$\frac{7}{4}, \frac{7}{5}, \frac{7}{6}, \frac{7}{7}$

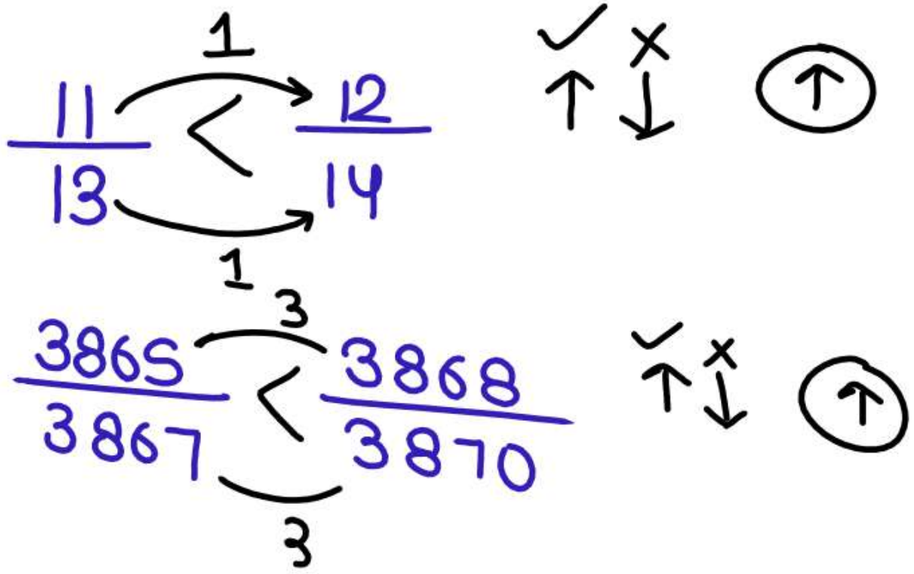
L S

$N^r \rightarrow$ Same

D↑ V↓

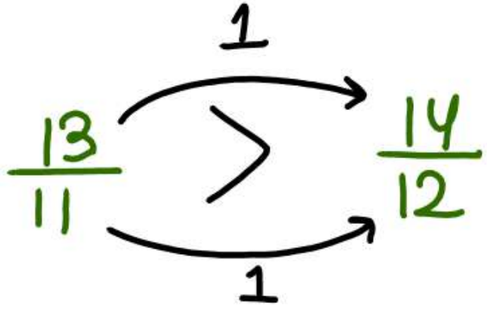

Final
N↑ D↑
i)
6/7
1
7/8
1
N↑ D↓
ii)
6/7
7/6
N↓ D↓
iii)
6/7
1
5/6
1
N↓ D↑
iv)
6/7
5/8

(i) N↑ D↑

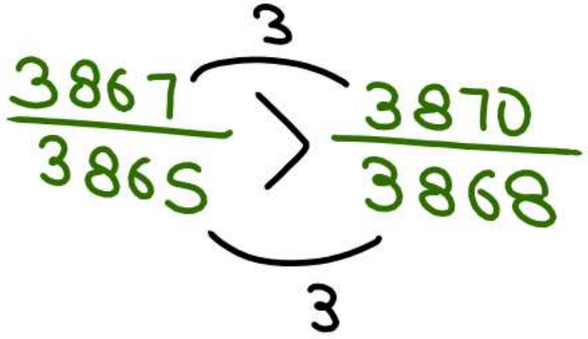

1
11
12
13
14
1
3
3865
3868
3867
3870
3

✓ ✗ ↑ ↓ (↑)

✓ ✗ ↑ ↓ (↑)


1
13
14
11
12
1

✗ ✓ ↑ ↓ ↓


3
3867
3870
3865
3868
3

✗ ✓ ↑ (↓) (↓)

(iii) N↓ D↓

$\frac{7}{8} > \frac{6}{7}$ (1 above, 1 below)

✓↓ ✗↑ ↓

$\frac{3312}{3313} > \frac{3311}{3312}$ (1 above, 1 below)

✓↓ ✗↑ ↓

$\frac{8}{7} < \frac{7}{6}$ (1 above, 1 below)

✗↓ ✓↑ ↑

$\frac{3313}{3312} < \frac{3312}{3311}$ (1 above, 1 below)

✗↓ ✓↑ ↑

(ii) N↑ D↓

$\frac{6}{11} < \frac{8}{9}$ ↑↑

$\frac{11}{9} < \frac{13}{8}$ ↑↑

$\frac{3865}{4717} < \frac{3870}{4716}$ (↑↑)

(iv) N↓ D↑

$\frac{6}{11} > \frac{4}{13}$ ↓↓

$\frac{9582}{13518} > \frac{8513}{17519}$ ↓↓

Unequal Diff.

N↑ D↑ ①

$\frac{10}{20} \overset{+5\ (50\%)}{\underset{+9\ (45\%)}{<}} \frac{15}{29}$

✓↑ ✗↓ Ⓣ

⊛

$\frac{4312}{5514} \overset{+40 \approx 1\%}{\underset{+100 \approx 2\%}{>}} \frac{4352}{5617}$

✗↑ ✓↓ ↓

Ladder

$$x = \cfrac{1}{1+\cfrac{1}{1+\cfrac{1}{3}}}$$

✓ $$x = 1+\cfrac{1}{1+\cfrac{1}{1+\cfrac{1}{3}}}$$

✓ $$x = 1-\cfrac{1}{1-\cfrac{1}{1-\cfrac{1}{3}}}$$

✓ $$x = 1+\cfrac{1}{1-\cfrac{1}{1+\cfrac{1}{1-\frac{1}{6}}}}$$

✓ $$x = 1+\cfrac{1}{2+\cfrac{1}{3-\frac{1}{4}}}$$

✓ $$x = 1+\cfrac{3}{2-\cfrac{3}{1-\frac{5}{6}}}$$

Ladder

$$x = \cfrac{1}{1+\cfrac{1}{1+\cfrac{1}{3}}}$$

$1 \pm 3 \overset{+1}{\underset{4}{\frown}} \quad \overset{+1}{\underset{7}{\frown}}$

$\text{ans} = \frac{4}{7}$

$$x = 1+\cfrac{1}{1+\cfrac{1}{1+\cfrac{1}{3}}}$$

$1 \pm 3 \overset{+1}{\underset{\pm 4}{\frown}} \times \overset{+1}{\underset{\pm 7}{\frown}} \times \overset{+1}{\underset{11}{\frown}}$

$\text{ans} = \frac{11}{7}$

Type-1

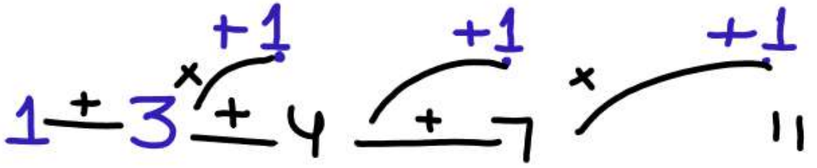
$$x = 1 + \cfrac{1}{1 + \cfrac{1}{1 + \cfrac{1}{3}}}$$

Ans

$$x = 1 + \cfrac{1}{1 + \cfrac{1}{\frac{4}{3}}}$$

$$= 1 + \cfrac{1}{1 + \frac{3}{4}}$$

$$= 1 + \cfrac{1}{\frac{7}{4}} = 1 + \frac{4}{7} = \frac{11}{7}$$



Ans = $\frac{11}{7}$

CONCEPT-05

(Ladder Fractions)

62. $1+\cfrac{1}{1+\cfrac{1}{1+\cfrac{4}{5}}}=?$

$4 \overset{+1}{\underset{+}{-}} 5 \overset{+1}{\underset{9}{\frown}} \overset{+1}{\underset{14}{}} \overset{+1}{\underset{23}{}}$

Ans $= \frac{23}{14}$

(a) $\frac{21}{17}$ (b) $\frac{17}{13}$

(c) $\frac{23}{14}$ (d) $\frac{23}{15}$

63. The value of $1+\cfrac{1}{1+\cfrac{1}{1+\cfrac{1}{1+\cfrac{1}{1+\cfrac{1}{1+\cfrac{2}{3}}}}}}$

+1 +1 +1 +1 +1

2 3 5 8 13 21 34

ans = $\frac{34}{21}$

(a) $\frac{21}{13}$

(b) $\frac{17}{2}$

(c) $\frac{34}{21}$

(d) $\frac{8}{5}$

64. **Find the value of x in the following equation:**

निम्नलिखित समीकरण में x का मान ज्ञात कीजिए:

$$\left[1+\cfrac{1}{1+\cfrac{1}{1+\cfrac{1}{5}}}\right] \div 1\frac{6}{11} = \frac{x}{2}$$

+1 +1 +1

1 — 5 | 6 | 11 | 17

$\frac{17}{11} \div \frac{17}{11} = \frac{x}{2}$

$2 = x$

NTPC CBT-2 17/06/2022 (Shift-3)

(a) 2

(b) 1

(c) $\frac{1}{2}$

(d) $\frac{2}{3}$

65. If $\cfrac{1}{1+\cfrac{1}{1+\cfrac{1}{1+\cfrac{1}{x}}}} = \frac{5}{8}$, then what is the value of x?

+1 +1 +1

1 x x+1 2x+1 3x+2

$\frac{2x+1}{3x+2} = \frac{5}{8}$

$\Rightarrow 16x + 8 = 15x + 10$

$\Rightarrow x = 2$

यदि $\cfrac{1}{1+\cfrac{1}{1+\cfrac{1}{1+\cfrac{1}{x}}}} = \frac{5}{8}$, हो, तो x का मान क्या है?

(a) 1
(b) 2
(c) 3
(d) 4

66. Find the value of $1-\cfrac{1}{1-\cfrac{1}{1-\cfrac{1}{1-\cfrac{2}{3}}}}$

2 — 3 ⌒ 1 (÷1) −2 (÷1) −3 (÷1) −1

ans = $\frac{+1}{+3}$

(a) $\frac{2}{3}$

(b) $-\frac{1}{3}$

(c) $\frac{1}{3}$

1-3 = -2
-2-1 = -3
-3-(-2)
-3+2 = -1

(d) $-\frac{2}{3}$

Q $x = 1 - \cfrac{1}{1 - \cfrac{1}{1 - \cfrac{1}{1 - \cfrac{1}{3}}}}$

ans 1 3 2 -1 -3 -2

$ans = \frac{+2}{73}$

$2 - 3 = -1$

$-1 - 2 = -3$

$-3 - (-1) = -3 + 1 = -2$

Q. $x = 1 + \cfrac{1}{1 - \cfrac{1}{1 + \cfrac{1}{1 + \cfrac{1}{1 - \cfrac{6}{7}}}}}$

Ans

6 7 −. 1 +. 8 +. 9 −. 1 +. 10

Ans = $\frac{10}{1}$

67. The value of the following is:

निम्नलिखित का मान है:

$$1-\cfrac{7}{1-\cfrac{1}{1+\cfrac{7}{1-\cfrac{7}{1}}}}$$

SSC CPO 27/06/2024 (Shift-02)

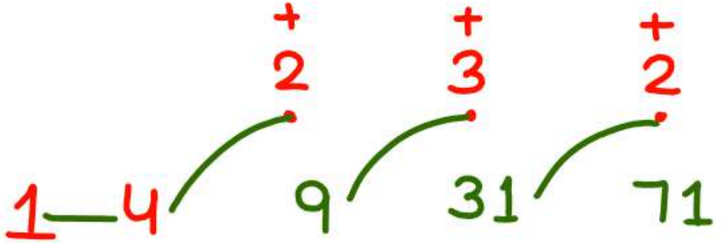
(a) 2
(b) 0
(c) 1
(d) –1

$-\frac{6}{7}+1=\frac{1}{7}$

$\frac{1}{7}-\left(\frac{-6}{7}\right)=\frac{1}{7}+\frac{6}{7}=\frac{7}{7}=1$

68. What will the value after simplifying this continued fraction?

इस निरंतर भिन्न को सरल बनाने के बाद मान क्या होगा?



ans = 31/71

$$\cfrac{1}{2+\cfrac{1}{3+\cfrac{1}{2+\cfrac{1}{4}}}}$$

NTPC CBT-1, 03/02/2021 (Shift-02)

31/71

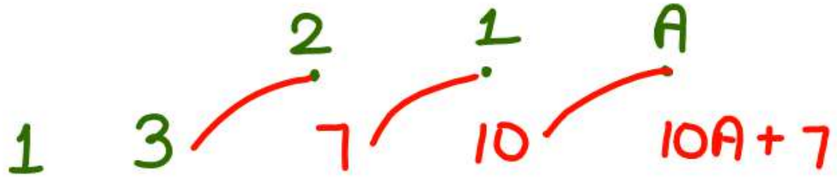
(a) 43/5

(b) 43/19

(c) 5/19

(d) 19/43

69. If $A+\cfrac{1}{1+\cfrac{1}{2+\cfrac{1}{3}}}=\cfrac{9}{10}$, then the value of A is:



$\frac{10A+7}{10}=\frac{9}{10}$

$\Rightarrow 10A=2$

$A=\frac{1}{5}$

यदि $A+\cfrac{1}{1+\cfrac{1}{2+\cfrac{1}{3}}}=\cfrac{9}{10}$, है, तो A का मान है:

SSC CPO 03/10/2023 (Shift-3)

(a) $\frac{3}{10}$

(b) $\frac{2}{5}$

(c) $\frac{1}{10}$

(d) $\frac{1}{5}$

T-4

① $x = 1 + \cfrac{1}{2 + \cfrac{1}{3 + \cfrac{1}{4 + \cfrac{6}{7}}}}$

$6 \xleftarrow{\times 4} 7 \xleftarrow{+} 34 \xleftarrow{\times 3} 109 \xleftarrow{+} 252 \xleftarrow{\times 2} 361 \xleftarrow{+} 1$

Ans: $\frac{361}{252}$

70. **Simplify:** $15+\cfrac{1}{6+\cfrac{1}{8+\cfrac{1}{10}}}$

8 6 15

1 10 — 81 496 7521

ans = $\frac{7521}{496}$

$= 15\frac{81}{496}$

CRPF HCM 01/03/2023 (Shift - 02)

(a) $15\frac{81}{496}$

(b) $15\frac{71}{186}$

(c) $15\frac{81}{472}$

(d) $15\frac{31}{374}$

71. $2+\cfrac{1}{2-\cfrac{1}{3-\cfrac{1}{4-\frac{1}{4}}}}=?$

$-4 \quad -3 \quad -2 \quad +2$

$1 \quad 4 \quad 15 \quad 41 \quad 67 \quad 175$

Ans $= \frac{175}{67}$

$= 2\frac{41}{67}$

CRPF HCM 28/02/2023 (Shift - 01)

(a) $\frac{41}{67}$

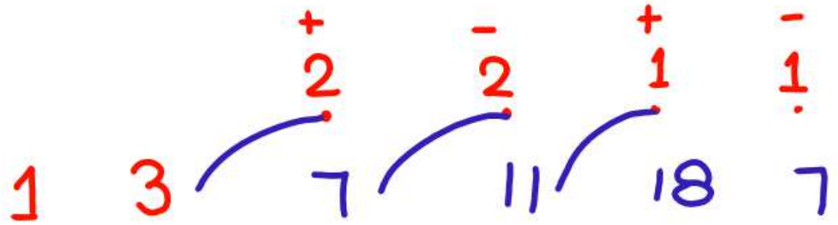
(b) $2\frac{15}{41}$

(c) $2\frac{41}{67}$ ✓

(d) $\frac{15}{41}$

72. The value of:/का मान है:

$$1-\cfrac{1}{1+\cfrac{1}{2-\cfrac{1}{2+\frac{1}{3}}}}$$



ans = $\frac{7}{18}$

SSC CPO 28/06/2024 (Shift-01)

(a) $\frac{7}{15}$

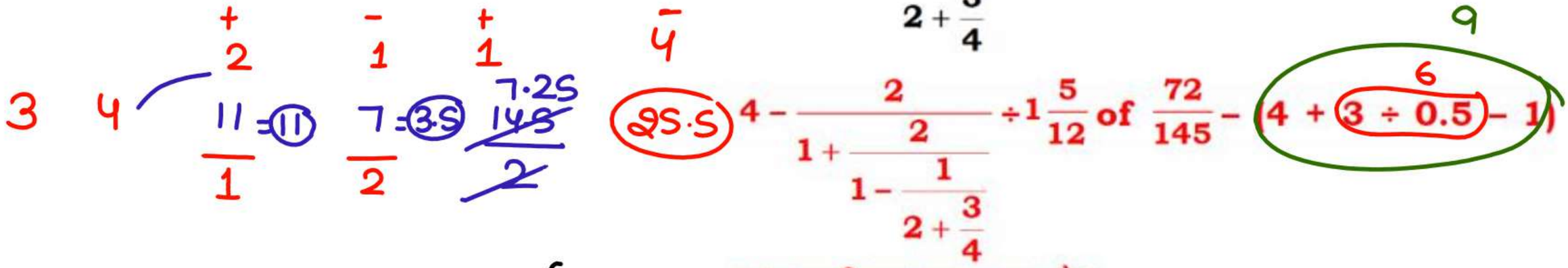
(b) $\frac{3}{16}$

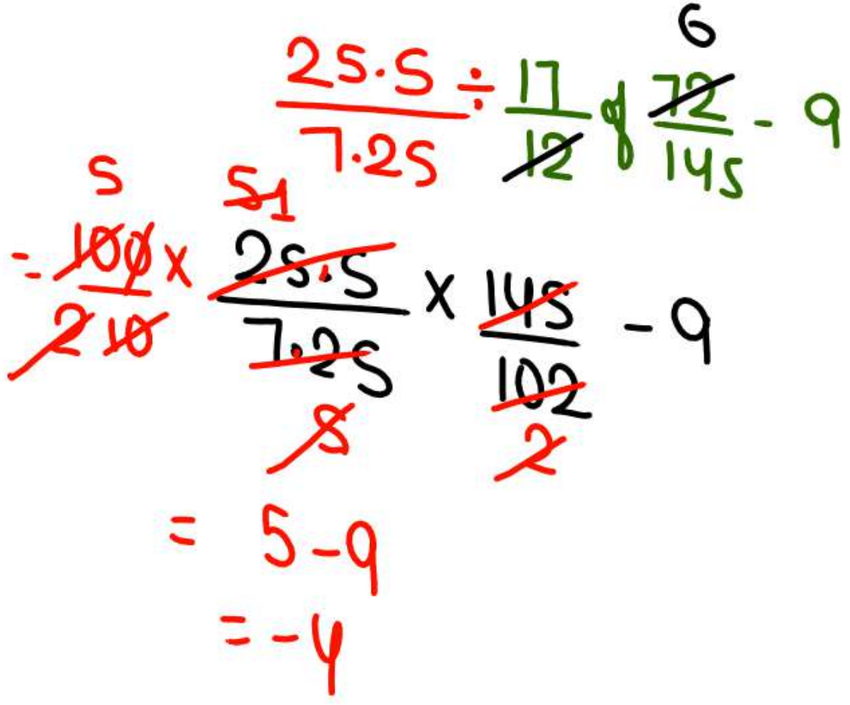
(c) $\frac{7}{18}$ ✓

(d) $\frac{3}{11}$

73. What is the simplified value of

$$4-\frac{2}{1+\frac{2}{1-\frac{1}{2+\frac{3}{4}}}}\div 1\frac{5}{12}\text{ of }\frac{72}{145}-(4+3\div 0.5-1)$$





का सरलीकृत मान क्या है?

SSC MTS 20/10/2021 (Shift-02)

(a) –4 ✓
(b) 1
(c) 3
(d) –2

= BODMAS

74. $3+\cfrac{5}{5+\cfrac{3}{1+\cfrac{7}{5}}}=?$

+1 +5 +3

7 ± 5 (12) (4) (25) (5) 19

ans = $\frac{19}{5}$

(a) $\frac{9}{13}$

(b) $\frac{13}{9}$

(c) $\frac{11}{2}$

(d) $\frac{19}{5}$

75. $\cfrac{1}{1+\cfrac{2}{1+\cfrac{3}{1+\cfrac{4}{1+\frac{4}{5}}}}}$ =?

4 5 9 ③ 8 ④ ⑦

+ 1 + 1 + 1

3 2

ans: $\frac{4}{7}$

(a) $\frac{7}{4}$

(b) $\frac{4}{7}$

(c) $\frac{11}{14}$

(d) $\frac{12}{5}$

M-1(Basic)

$$\frac{13}{29} = \frac{1}{\frac{29}{13}}$$

$$= \frac{1}{2+\frac{3}{13}} = \frac{1}{2+\frac{1}{\frac{13}{3}}}$$

$$= \frac{1}{2+\frac{1}{4+\frac{1}{3}}} = \frac{1}{a+\frac{1}{b+\frac{1}{c}}}$$

a : 2
b : 4
c : 3

a - b + c
2 - 4 + 3
= 1

76. $\dfrac{1}{a+\dfrac{1}{b+\dfrac{1}{c}}} = \dfrac{13}{29}$ **find a – b + c=?**

(a) 1 **(b) 2**

(c) 0 **(d) 3**

$\frac{29}{13} = 2 + \frac{3}{13}$

$\frac{13}{3} = 4 + \frac{1}{3}$

76. $\cfrac{1}{a + \cfrac{1}{b + \cfrac{1}{c}}} = \frac{13}{29}$ **find a – b + c=?**

$a - b + c$
$= 2 - 4 + 3$
$= 1$

(a) 1 (b) 2
(c) 0 (d) 3

77. $$\cfrac{1}{a+\cfrac{1}{b+\cfrac{1}{c+\cfrac{1}{d}}}} = \frac{17}{60}$$ find a + b + c + d =?

$\frac{60}{17} = 3 + \frac{9}{17}$

$\frac{17}{9} = 1 + \frac{8}{9}$

$\frac{9}{8} = 1 + \frac{1}{8}$

(a) 11 (b) 12

(c) 13 (d) 14

$a+b+c+d$

$= 3+1+1+8$

$= 13$

78. $$\cfrac{1}{a+\cfrac{1}{2b+\cfrac{1}{3c+\cfrac{1}{d}}}} = \frac{30}{43}$$ find a + b + c + d = ?

(a) 10 (b) 11

(c) ~~7~~ (d) None

$\frac{43}{30} = 1 + \frac{13}{30}$

$\frac{30}{13} = 2 + \frac{4}{13}$

$\frac{13}{4} = 3 + \frac{1}{4}$

$a = 1$

$2b = 2\ 1$

$3c = 3\ 1$

$d = 4$

$a+b+c+d = 7$

79. If a, b, c, d are integers such that

H.W

$$\frac{1}{a+\dfrac{1}{b+\dfrac{1}{c+\dfrac{1}{d}}}} = \frac{29}{154}$$, then a + b + c + d = ?

(a) 12 (b) 13

(c) 14 (d) 15

80. If $\frac{45}{53} = \cfrac{1}{a + \cfrac{1}{b + \cfrac{1}{c - \frac{2}{5}}}}$, where a, b and c are Positive integers, then what is the value of (4a – b + 3c)?

SSC CGL TIER - II 15/11/2020

(a) 5 (b) 4

(c) 6 (d) 7

$\frac{53}{45} = \overset{a}{1} + \frac{8}{45}$

$\frac{45}{8} = \overset{b}{5} + \frac{5}{8}$

$\frac{8}{5} = \overset{c}{2} - \frac{2}{5}$

$4a - b + 3c$

$= 4 - 5 + 6$

$= 5$

$\frac{8}{5} \rightarrow 1 + \frac{3}{5}$

$\rightarrow 2 \; -\frac{2}{5}$

81. If $\cfrac{1}{x+\cfrac{1}{y+\cfrac{2}{z+\cfrac{1}{4}}}} = \cfrac{29}{79}$, where x, y and y are natural numbers, then the value $(2x + 3y - z)$ is:

SSC CGL TIER - II 16/11/2020

(a) 1 (b) 4

(c) 0 (d) 2

$$\frac{29}{79} = \frac{1}{\frac{79}{29}} = \frac{1}{2+\frac{21}{29}}$$

$$\frac{1}{2+\frac{1}{\frac{29}{21}}} = \frac{1}{2+\frac{1}{1+\frac{8}{21}}}$$

$$= \frac{1}{2+\frac{1}{1+\frac{2}{\frac{21}{4}}}} = \frac{1}{\underset{x}{2}+\frac{1}{\underset{y}{1}+\frac{2}{\underset{z}{5}+\frac{1}{4}}}}$$

$$2x+3y-z$$
$$= 4+3-5$$
$$= 2$$

81. If $\cfrac{1}{x+\cfrac{1}{y+\cfrac{2}{z+\cfrac{1}{4}}}} = \cfrac{29}{79}$, where x, y and y are natural numbers, then the value $(2x + 3y - z)$ is:

SSC CGL TIER - II 16/11/2020

(a) 1 (b) 4

(c) 0 (d) 2

$\frac{79}{29} = 2 + \frac{21}{29}$ (x)

$\frac{29}{21} = 1 + \frac{8}{21} = \frac{2 \times 4}{21}$ (y)

$\frac{21}{4} = 5 + \frac{1}{4}$ (z)

$2x + 3y - z$

$= 4 + 3 - 5$

$= 2$

CONCEPT-06

(Continuous Fraction Series
सतत भिन्नात्मक श्रेणी)

1) $\frac{1}{1\times2}+\frac{1}{2\times3}+\frac{1}{3\times4}+\cdots$

2) $\frac{1}{1\times2\times3}+\frac{1}{2\times3\times4}+\frac{1}{3\times4\times5}+\cdots$

3) $\frac{1}{1\times2\times3\times4}+\frac{1}{2\times3\times4\times5}+\cdots$

4) $\frac{1}{2^2-1^2}+\frac{1}{3^2-2^2}+\cdots$

82. $\left[\left(1-\frac{1}{3}\right)\left(1-\frac{1}{4}\right)\left(1-\frac{1}{5}\right)\left(1-\frac{1}{6}\right)....\left(1-\frac{1}{99}\right)\left(1-\frac{1}{100}\right)\right]$

(a) $\frac{2}{99}$

(b) $\frac{1}{25}$

(c) $\frac{1}{50}$

(d) $\frac{1}{100}$

$\frac{2}{3} \times \frac{3}{4} \times \frac{4}{5} \times \frac{5}{6} \times \frac{6}{7} \cdots \frac{98}{99} \times \frac{99}{100}$

$= \frac{2}{100} = \frac{1}{50}$

Concept of Series

$$\frac{1}{1\times2} + \frac{1}{2\times3} + \frac{1}{3\times4} + \frac{1}{4\times5}$$

Basic

$$\frac{1}{1} - \cancel{\frac{1}{2}} + \cancel{\frac{1}{2}} - \cancel{\frac{1}{3}} + \cancel{\frac{1}{3}} - \cancel{\frac{1}{4}} + \cancel{\frac{1}{4}} - \frac{1}{5}$$

$$= \frac{1}{1} - \frac{1}{5}$$

$$= \frac{4}{5}$$

Trick:

$$\frac{1}{\text{diff}}\left[\frac{1}{1^{st}} - \frac{1}{\text{last}}\right]$$

$$\rightarrow \frac{1}{1}\left[\frac{1}{1} - \frac{1}{5}\right]$$

$$= 1\left[\frac{4}{5}\right]$$

$$= \frac{4}{5}$$

$n=1$

83. **The sum of**

$$\frac{1}{2}+\frac{1}{6}+\frac{1}{12}+\ldots\ldots+\frac{1}{n(n+1)}$$ **is:**

$\frac{1}{2}+\frac{1}{6}+\frac{1}{12}+\ldots\ldots+\frac{1}{n(n+1)}$ का योग है-

NTPC CBT-1, 02/03/2021 (Shift-03)

(a) $\left(\frac{n+1}{n}\right)\frac{2}{1}$

(b) $\frac{n+1}{2n}\ \frac{2}{2}$ 1

(c) $\frac{n(n+1)}{2}\ \frac{2}{2}$ 1

(d) ✓ $\frac{n}{n+1}\ \frac{1}{2}$

83. **The sum of**

$$\frac{1}{2}+\frac{1}{6}+\frac{1}{12}+\ldots..+\frac{1}{n(n+1)}$$ **is:**

$$\frac{1}{2}+\frac{1}{6}+\frac{1}{12}+\ldots..+\frac{1}{n(n+1)}$$ का योग है-

NTPC CBT-1, 02/03/2021 (Shift-03)

(a) $\left(\frac{n+1}{n}\right)$

(b) $\frac{n+1}{2n}$

(c) $\frac{n(n+1)}{2}$

(d) $\frac{n}{n+1}$

$\frac{1}{1\times2}+\frac{1}{2\times3}+\frac{1}{3\times4}+\ldots..+\frac{1}{n\times(n+1)}$

$=\frac{1}{1}\left[\frac{1}{1}-\frac{1}{(n+1)}\right]$

$=1\left[\frac{(n+1)-1}{(n+1)}\right]$

$=\frac{n}{n+1}$

84. $\frac{1}{11\times12}+\frac{1}{12\times13}+\frac{1}{13\times14}+\ldots\ldots+\frac{1}{80\times81}$

(a) $\frac{69}{890}$

(b) $\frac{70}{891}$

(c) $\frac{71}{790}$

(d) $\frac{72}{891}$

Ans = $\frac{1}{1}\left[\frac{1}{11}-\frac{1}{81}\right]$

$= \frac{81-11}{11\times81} = \frac{70}{891}$

85. $\frac{1}{1\times 4}+\frac{1}{4\times 7}+\frac{1}{7\times 10}+\ldots\ldots+\frac{1}{97\times 100}$

(a) $\frac{33}{100}$ (b) $\frac{34}{99}$

(c) $\frac{35}{99}$ (d) $\frac{37}{100}$

$= \frac{1}{3}\left[\frac{1}{1}-\frac{1}{100}\right]$

$= \frac{1}{3}\times\frac{99}{100}33 = \frac{33}{100}$

86. $\frac{1}{3\times 7}+\frac{1}{7\times 11}+\frac{1}{11\times 15}+\ldots.+\frac{1}{899\times 903}$

(a) $\frac{21}{509}$

(b) $\frac{18}{409}$

~~(c)~~ $\frac{25}{301}$

(d) $\frac{29}{31}$

Ans: $\frac{1}{4}\left[\frac{1}{3}-\frac{1}{903}\right]$

$= \frac{1}{4}\times\frac{900}{3\times 903}\times\frac{75}{301}=\frac{25}{301}$

86. $\frac{1}{3\times 7}+\frac{1}{7\times 11}+\frac{1}{11\times 15}+\ldots.+\frac{1}{899\times 903}$

(a) $\frac{21}{509}$

(b) $\frac{18}{409}$

(c) $\frac{25}{301}$

(d) $\frac{29}{31}$

Trick $\frac{1}{3\times 903}=\frac{1}{3\times 3\times 301}$

87. $\frac{1}{4\times 9}+\frac{1}{9\times 14}+\frac{1}{14\times 19}+\ldots\ldots+\frac{1}{99\times 104}$

Try

(a) $\frac{7}{104}$

(b) $\frac{9}{100}$

(c) $\frac{5}{104}$

(d) $\frac{8}{105}$

Q. $\frac{1}{1\times2\times3} + \frac{1}{3\times4\times5} + - - - + \frac{1}{9\times10\times11}$

~~Ans~~

$$ans = \frac{1}{diff}\left[\frac{1}{1^{st}} - \frac{1}{last}\right]$$

$$= \frac{1}{2}\left[\frac{1}{1\times2} - \frac{1}{10\times11}\right]$$

$$= \frac{1}{2}\left(\frac{1}{2} - \frac{1}{110}\right)$$

$$= \frac{1}{2} \times \frac{108}{2\times110}^{54} = \frac{27}{110}$$

88. $\frac{1}{1\times2\times3}+\frac{1}{2\times3\times4}+\frac{1}{3\times4\times5}+\ldots.+\frac{1}{98\times99\times100}$

(a) $\frac{4949}{19800}$ (b) $\frac{1980}{49490}$

(c) $\frac{9898}{19800}$ (d) $\frac{1980}{47490}$

$= \frac{1}{2}\left[\frac{1}{1\times2}-\frac{1}{99\times100}\right]$

$= \frac{1}{2}\left(\frac{1}{2}-\frac{1}{9900}\right)$

$= \frac{1}{2}\times\frac{9898}{2\times9900}$ 4949

$= \frac{4949}{19800}$

$\frac{1}{\text{diff}}\left[\frac{1}{1\text{st}}-\frac{1}{\text{last}}\right]$

89. Which of the following statement is true?

I. $\frac{1}{2}+\frac{1}{6}+\frac{1}{12}+\ldots\ldots\ldots\frac{1}{110}<\frac{5}{6}$

II. $\frac{1}{3}+\frac{1}{15}+\frac{1}{35}+\ldots\ldots\ldots\frac{1}{143}>\frac{7}{13}$

SSC CHSL 13/03/2023 (Shift-04)

(a) Only I
(b) Both I and II
(c) Only II
(d) Neither I nor II

I $\frac{1}{1\times2}+\frac{1}{2\times3}+\ldots+\frac{1}{10\times11}$

$=\frac{1}{1}\left[\frac{1}{1}-\frac{1}{11}\right]$

$=\frac{10}{11}<\frac{5}{6}$

II $\frac{1}{1\times3}+\frac{1}{3\times5}+\ldots+\frac{1}{11\times13}$

$=\frac{1}{2}\left[\frac{1}{1}-\frac{1}{13}\right]$

$=\frac{1}{2}\times\frac{12}{13}=\frac{6}{13}>\frac{7}{13}$

$x = \frac{1}{1}\left[\frac{1}{12} - \frac{1}{24}\right]$

$= 1 \times \frac{1}{24} = \frac{1}{24}$

$y = \frac{1}{1}\left[\frac{1}{36} - \frac{1}{72}\right] = \frac{1}{72}$

$\frac{x}{y} = \frac{\frac{1}{24}}{\frac{1}{72}} = \frac{72}{24}$ ③

90. If $x = \frac{1}{12.13} + \frac{1}{13.14} + \frac{1}{14.15} \ldots\ldots + \frac{1}{23.24}$, $y = \frac{1}{36.37} + \frac{1}{37.38} + \frac{1}{38.39} \ldots\ldots + \frac{1}{71.72}$ then $\frac{x}{y}$ is equal to:

(a) $\frac{1}{3}$

(b) $\frac{1}{24}$

(c) $\frac{1}{72}$

(d) 3

91. $\dfrac{1}{1\times3\times5}+\dfrac{1}{3\times5\times7}+\ldots\ldots+\dfrac{1}{9\times11\times13}$

$\frac{1}{4}\left[\frac{1}{1\times3}-\frac{1}{11\times13}\right]$

$=\frac{1}{4}\left(\frac{1}{3}-\frac{1}{143}\right)$

$=\frac{1}{4}\times\frac{140}{3\times143}35=\frac{35}{429}$

(a) $\dfrac{35}{429}$

(b) $\dfrac{35}{439}$

(c) $\dfrac{25}{329}$

(d) $\dfrac{25}{329}$

91. $\frac{1}{1\times3\times5}+\frac{1}{3\times5\times7}+\ldots\ldots+\frac{1}{9\times11\times13}$

(a) $\frac{35}{429}$

(b) $\frac{35}{439}$

(c) $\frac{25}{329}$

(d) $\frac{25}{329}$

Trick. $\frac{1}{3\times11\times13}$

92. $\frac{1}{1\times2\times3\times4}+\frac{1}{2\times3\times4\times5}+\ldots\ldots+\frac{1}{6\times7\times8\times9}$

(a) $\frac{83}{1512}$ ✓

(b) $\frac{84}{1513}$

(c) $\frac{83}{1415}$

(d) $\frac{84}{1413}$

$= \frac{1}{3}\left[\frac{1}{1\times2\times3}-\frac{1}{7\times8\times9}\right]$

$= \frac{1}{3}\left[\frac{1}{6}-\frac{1}{504}\right]$

$= \frac{1}{3}\times\frac{498}{6\times504}\,83$

$= \frac{83}{1512}$

$\frac{1}{\text{diff}}\left[\frac{1}{1^{st}}-\frac{1}{last}\right]$

92. $\frac{1}{1\times2\times3\times4}+\frac{1}{2\times3\times4\times5}+\ldots\ldots+\frac{1}{6\times7\times8\times9}$

(a) $\frac{83}{1512}$

(b) $\frac{84}{1513}$

(c) $\frac{83}{1415}$

(d) $\frac{84}{1413}$

Trick $\frac{1}{6\times7\times8\times9}$

$\frac{1}{6}\left(\frac{1}{1\times3\times5} - \frac{1}{13\times15\times17}\right)$

93. $\frac{1}{1\times3\times5\times7} + \frac{1}{3\times5\times7\times9} + \ldots\ldots + \frac{1}{11\times13\times15\times17}$

$= \frac{1}{6}\left(\frac{1}{15} - \frac{1}{3315}\right)$

$= \frac{1}{6} \times \frac{220}{3300} \Big/ \frac{3}{18\times3315}$ $\quad 663$

$= \frac{22}{1989}$

(a) $\frac{20}{1991}$ (b) $\frac{22}{1989}$

(c) $\frac{25}{1990}$ (d) $\frac{27}{1991}$

93. $\frac{1}{1\times3\times5\times7}+\frac{1}{3\times5\times7\times9}+\ldots\ldots+\frac{1}{11\times13\times15\times17}$

(a) $\frac{20}{1991}$

(b) $\frac{22}{1989}$

(c) $\frac{25}{1990}$

(d) $\frac{27}{1991}$

Trick: $\frac{1}{15\times13\times15\times17}$

94. $\frac{1}{4\times11\times18}+\frac{1}{11\times18\times25}+\frac{1}{18\times25\times32}+\ldots\ldots+\frac{1}{67\times74\times81}$

Try

(a) $\frac{425}{263736}$

(b) $\frac{425}{253737}$

(c) $\frac{424}{253737}$

(d) $\frac{425}{253736}$

Trick

$$\frac{1}{1\times2\times3\times4\times11\times12\times13\times14}$$

95. $\dfrac{1}{1\times2\times3\times4\times5} + \dfrac{1}{2\times3\times4\times5\times6} + \ldots\ldots\ldots + \dfrac{1}{10\times11\times12\times13\times14}$

(a) $\dfrac{10009}{960960}$ ✓

(b) $\dfrac{10009}{960970}$

(c) $\dfrac{10019}{960961}$

(d) $\dfrac{10018}{960961}$

Q $\frac{1}{1\times4} + \frac{1}{4\times7} + \frac{1}{7\times10}$

Ans $\frac{1}{\text{diff}}\left[\frac{1}{1\text{st}} - \frac{1}{\text{last}}\right]$

Q $\frac{3}{1\times4} + \frac{3}{4\times7} + \frac{3}{7\times10}$

Ans $3\left[\frac{1}{1\times4} + \frac{1}{4\times7} + \frac{1}{7\times10}\right]$

$= \cancel{3} \times \frac{1}{\cancel{3}}\left[\frac{1}{1\text{st}} - \frac{1}{\text{last}}\right]$

$= \left[\frac{1}{1\text{st}} - \frac{1}{\text{last}}\right]$

$\frac{5}{4\times9} + \frac{7}{9\times16} + \ldots + \frac{19}{81\times100}$

$= \left[\frac{1}{1^{st}} - \frac{1}{last}\right]$

$= \left[\frac{1}{4} - \frac{1}{100}\right]$

$= \frac{24}{100} \quad 6$

25

96. $\frac{5}{2^2.3^2} + \frac{7}{3^2.4^2} + \frac{9}{4^2.5^2} + \frac{11}{5^2.6^2} + \frac{13}{6^2.7^2} + \frac{15}{7^2.8^2} + \frac{17}{8^2.9^2} + \frac{19}{9^2.10^2}$ **is equal to.**

(a) $\frac{1}{100}$

(b) $\frac{6}{25}$

(c) $\frac{101}{100}$

(d) 1

97. $\frac{4}{3\times 7}+\frac{6}{7\times 13}+\frac{8}{13\times 21}+\frac{10}{21\times 31}+\frac{12}{31\times 43}$

(a) $\frac{39}{128}$ (b) $\frac{40}{129}$

(c) $\frac{41}{130}$ (d) $\frac{42}{135}$

$= \frac{1}{3}-\frac{1}{43}$

$= \frac{43-3}{3\times 43} = \frac{40}{129}$

$$\frac{40}{483} + \frac{10}{31}$$

$$= \frac{1240 + 4830}{483 \times 31}$$

$$= \frac{6070}{14973}$$

98. $\frac{1}{1\times3\times5} + \frac{1}{1\times4} + \frac{1}{3\times5\times7} + \frac{1}{4\times7} + \frac{1}{5\times7\times9} + \frac{1}{7\times10} + \ldots + \text{upto 20 terms}$

(a) $\frac{6179}{15275}$

(b) $\frac{6070}{14973}$

(c) $\frac{7191}{15174}$

(d) $\frac{5183}{16423}$

$$\underbrace{\left(\frac{1}{1\times3\times5}+\frac{1}{3\times5\times7}+\cdots+\frac{1}{a_{10}}\right)}_{10\text{ Terms}}+\underbrace{\left(\frac{1}{1\times4}+\frac{1}{4\times7}+\cdots+\frac{1}{a_{10}}\right)}_{10\text{ Terms}}$$

$1, 3, \ldots\ldots a_{10}$

$a_{10} = a + 9d$

$= 1 + 18$

$= 19$

$1, 4, \ldots\ldots a_{10}$

$a_{10} = a + 9d$

$= 1 + 27 = 28$

$$\left[\frac{1}{1\times3\times5}+\frac{1}{3\times5\times7}+\cdots+\frac{1}{19\times21\times23}\right]+\left[\frac{1}{1\times4}+\frac{1}{4\times7}+\cdots+\frac{1}{28\times31}\right]$$

$$=\frac{1}{4}\left[\frac{1}{1\times3}-\frac{1}{21\times23}\right]+\frac{1}{3}\left[\frac{1}{1}-\frac{1}{31}\right]$$

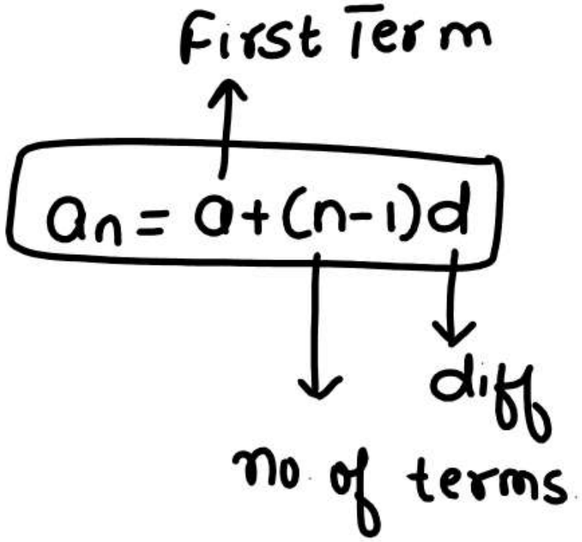
$$=\frac{1}{\cancel{4}}\times\frac{\cancel{80}\,40}{\cancel{3}\times21\times23}+\frac{1}{\cancel{3}}\times\frac{\cancel{30}\,10}{31}$$

① 2, 5, 8, 11 ..... (a = 2, d)

$a_{20} = a + 19d = 2 + 57 = 59$

$a_{25} = a + 24d = 2 + 72 = 74$

$a_{50} = a + 49d = 2 + 147 = 149$



99. $a_1 = \frac{1}{2\times5}$, $a_2 = \frac{1}{5\times8}$, $a_3 = \frac{1}{8\times11}$

then, $a_1 + a_2 + ... + a_{100} + ?$

(a) $\frac{25}{151}$ **(b)** $\frac{30}{157}$

(c) $\frac{1}{4}$ **(d)** $\frac{9}{55}$

$\frac{1}{2\times5} + \frac{1}{5\times8} + \frac{1}{8\times11} + \cdots + \frac{1}{299\times302}$

$= \frac{1}{3}\left[\frac{1}{2} - \frac{1}{302}\right]$

$= \frac{1}{3} \times \frac{300}{604} \frac{100}{} \frac{25}{151}$

$2, 5, 8 \ldots\ldots a_{100}$

$a_{100} = a + 99d$

$= 2 + 297$

$= 299$

100. $\frac{1}{2^2-1^2}+\frac{1}{4^2-1}+\frac{1}{6^2-1}+\ldots..+\frac{1}{20^2-1}$

(a) $\frac{9}{19}$ (b) $\frac{10}{19}$

(c) $\frac{11}{19}$ (d) $\frac{10}{21}$

$\frac{1}{1\times3}+\frac{1}{3\times5}+\frac{1}{5\times7}+\cdots\cdots+\frac{1}{19\times21}$

$=\frac{1}{2}\left[\frac{1}{1}-\frac{1}{21}\right]$

$=\frac{1}{2}\times\frac{20}{21}10=\frac{10}{21}$

101. $\frac{1}{7^2-3^2}+\frac{1}{13^2-3^2}+\frac{1}{19^2-3^2}+\ldots.+\frac{1}{49^2-3^2}$

(a) $\frac{1}{26}$ ✓ (b) $\frac{3}{52}$

(c) $\frac{1}{13}$ (d) $\frac{3}{26}$

$= \frac{1}{4\times10}+\frac{1}{10\times16}+\ldots\ldots+\frac{1}{46\times52}$

$= \frac{1}{6}\left[\frac{1}{4}-\frac{1}{52}\right]$

$= \frac{1}{6}\times\frac{12}{52}_{26} = \frac{1}{26}$

CONCEPT-07

(a $\frac{b}{c}$ Types of numbers

प्रकार की संख्याएं)

(a) If denominator of a number same as multiplier

यदि किसी संख्या का हर उसके गुणज के समान हो तो

102. The value of $99\frac{95}{99} \times 99$ is

(a) 9798 (b) 9997

(c) 9898 (d) 9896

Basic

$\left[99\frac{95}{99}\right] \times 99$

$= \left[99 + \frac{95}{99}\right] \times 99$

$= 99 \times 99 + \frac{95}{99} \times 99$

$= (100-1) \times 99 + 95$

$= 9900 - 99 + 95$

$= 9900 - 4$

$= 9896$

$2\frac{3}{4} = 2 + \frac{3}{4}$

$1\frac{7}{8} = 1 + \frac{7}{8}$

102. The value of $99\frac{95}{99} \times 99$ is

(a) 9798 (b) 9997

(c) 9898 (d) 9896

Basic

$$\left[99\frac{95}{99}\right] \times 99$$

$$= \left[99 + \frac{95}{99}\right] \times 99$$

$$= 99 \times 99 + \frac{95}{99} \times 99$$

$$= 9801 + 95$$

$$= 9896$$

$2\frac{3}{4} = 2 + \frac{3}{4}$

$1\frac{7}{8} = 1 + \frac{7}{8}$

$$\begin{array}{r} 99 \\ \times 23 \\ \hline 2277 \end{array}$$

$$\begin{array}{r} 99 \\ \times 87 \\ \hline 8613 \end{array}$$

$$\begin{array}{r} 99 \\ \times 93 \\ \hline 9207 \end{array}$$

$$\begin{array}{r} 99 \\ \times 76 \\ \hline 7524 \end{array}$$

$$\begin{array}{r} 99 \\ \times 64 \\ \hline 6336 \end{array}$$

$$\begin{array}{r} 99 \\ \times 57 \\ \hline 5643 \end{array}$$

102. The value of $99\frac{95}{99} \times 99$ is

(a) 9798 (b) 9997

(c) 9898 (d) 9896

Trick:

$9900 - 4$

$= 9896$

* $99\frac{93}{99} \times 99 = 9900 - 6 = 9894$

* $999\frac{87}{99} \times 99 = 99000 - 12 = 98988$

* $9999\frac{873}{999} \times 999 = 9990000 - 126$

$= 9989874$

* $99\frac{85}{89} \times 89$

$= \left(99 + \frac{85}{89}\right) \times 89$

$= 99 \times 89 + 85$

$= (100-1) \times 89 + 85$

$= 8900 - 89 + 85$

$= 8900 - 4$

$= \underline{8896}$

* $99\ \frac{85}{89} \times 89$

$\underline{\text{ans}} = 8900 - 4$

$= 8896$

* $9\frac{894}{899} \times 899 = 8990 - 5 = 8985$

* $99\frac{763}{799} \times 799 = 79900 - 36 = 79864$

* $99\frac{683}{685} \times 685 = 68500 - 2 = 68498$

* $999\frac{57}{63} \times 63 = 63000 - 6 = 62994$

103. $999\frac{98}{99} \times 99$ is equal to:

(a) 98999 (b) 99899

(c) 99989 (d) 99998

$$= 99000 - 1$$
$$= 98999$$

104. $999\frac{994}{999} \times 999$

(a) 908999 (b) 999099

(c) 998995 (d) 989095

$= 999000 - 5$

$= 998995$

(b) If difference between numerator and denominator is 1.

यदि अंश और हर के बीच का अंतर 1 हो तो

105. $\frac{1}{8} + 999\frac{791}{792} \times 99$

(a) 89000 (b) 88900

(c) 95900 (d) 99000

Basic

$\frac{1}{8} + \left(999 + \frac{791}{792}\right) \times 99$

$= \frac{1}{8} + 999 \times 99 + \frac{791}{792} \times 99$

$= \frac{1}{8} + \frac{791}{8} + 99 \times (1000 - 1)$

$= 99\frac{792}{8} + 99000 - 99$

$= 99000 + 99 - 99$

105. $\frac{1}{8} + 999\frac{791}{792} \times 99$

(a) 89000 (b) 88900

(c) 95900 (d) 99000 ✓

Basic

$$\frac{1}{8} + \left(999 + \frac{791}{792}\right) \times 99$$

$$= \frac{1}{8} + 999 \times 99 + \frac{791}{792} \times 99$$

$$= \frac{1}{8} + \frac{791}{8} + 99 \times (1000 - 1)$$

$$= 99\frac{792}{8} + 99000 - 99$$

$$= 99000 + 99 - 99$$

Trick

99000

106. Find the value of $\frac{1}{5} + 999\frac{494}{495} \times 99$

(a) 90000 (b) 99000

(c) 90900 (d) 99990

Basic

$$\frac{1}{5} + \left(999 + \frac{494}{495}\right) \times 99$$

$$= \frac{1}{5} + 99 \times (999) + \frac{494}{5}$$

$$= \frac{495}{5} + 99(1000 - 1)$$

$$= 99 + 99000 - 99$$

Trick

99000

107. $\frac{1}{7}+\left(999\frac{692}{693}\right)\times 99$ is equal to:

(a) 1

(b) 99000

(c) 99800

(d) 99900

ans = 99000

Basic

(c) Series Type/श्रेणी प्रकार

$999+\frac{1}{5}+999+\frac{2}{5}+999+\frac{3}{5}+999+\frac{4}{5}$

108. $999\frac{1}{5}+999\frac{2}{5}+\ldots\ldots\ldots+999\frac{4}{5}$

(a) 3798 (b) 3998

(c) 3899 (d) 9939

$= 999\times4+\left(\frac{1}{5}+\frac{2}{5}+\frac{3}{5}+\frac{4}{5}\right)$

$= 3996+\frac{10}{5}2$

$= 3998$

(c) Series Type/श्रेणी प्रकार

108. $999\frac{1}{5} + 999\frac{2}{5} + \ldots\ldots 999\frac{3}{5} \ldots\ldots + 999\frac{4}{5}$

(a) 3798 (b) 3998

(c) 3899 (d) 9939

$999 \times 4 + \frac{10^2}{5}$

$= 3998$

1, 2, 3, 4

109. Find the value of

$$777\frac{1}{5}+777\frac{2}{5}+777\frac{3}{5}+777\frac{4}{5}$$

$777\frac{1}{5}+777\frac{2}{5}+777\frac{3}{5}+777\frac{4}{5}$ का मान ज्ञात कीजिए।

NTPC CBT-1, 03/03/2021 (Shift-01)

(a) 3110
(b) 3018
(c) 3000
(d) 3108

$777\times4+\frac{10}{5}\times2$

$=3108+2$

$=3110$

CONCEPT-08

(Some Standard Formulae for Addition

जोड़ के लिए कुछ सामान्य सूत्र)

- **Sum of first 'n' natural numbers**

  प्रथम **n** प्राकृतिक संख्या का योगफल

$$1 + 2 + 3 + \ldots\ldots\ldots + n = \frac{n(n+1)}{2}$$

- **Sum of the squares of first 'n' natural numbers.**

  प्रथम **n** प्राकृतिक संख्याओं के वर्गों का योगफल

$$1^2 + 2^2 + 3^2 + \ldots\ldots\ldots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

- **Sum of square of n odd/even number/n** विषम/सम संख्याओं के वर्गों का योग = $\frac{n(n+1)(n+2)}{6}$ **where n is last odd/even number/**जहाँ **n** अंतिम विषम/सम संख्या है।

- **Sum of the cubes of first 'n' natural numbers.**

  प्रथम n प्राकृतिक संख्याओं के घनों का योगफल

$$1^3 + 2^3 + 3^3 + \ldots\ldots\ldots + n^3 = \left\{ n\frac{(n+1)}{2} \right\}^2$$

- **Sum of even integers/**सम पूर्णांकों का योगफल

  $2 + 4 + 6 \ldots\ldots\ldots + 2n = n(n + 1)$

- **Sum of odd integers/**विषम पूर्णांकों का योगफल

  $1 + 3 + 5 + \ldots\ldots\ldots\ldots (2n - 1) = n^2$

* $1+2+3+4+\ldots\ldots+n = \frac{n(n+1)}{2}$

* $1^2+2^2+3^2+4^2+\ldots\ldots+n^2 = \frac{n(n+1)(2n+1)}{6}$

* $1^3+2^3+3^3+4^3+\ldots\ldots+n^3 = \left[\frac{n(n+1)}{2}\right]^2$

* $2+4+6+8+10+\ldots\ldots+n = n(n+1)$

* $1+3+5+7+9+\ldots\ldots+n = n^2$

$n \rightarrow$ first n consecutive no

Natural No

Q $1+2+3+4+\ldots+10 = \frac{n(n+1)}{2} = \frac{\overset{5}{\cancel{10}} \times 11}{\cancel{2}} = 55$

Q $1+2+3+4+\ldots+50 = \;''\; = \frac{\overset{25}{\cancel{50}} \times 51}{\cancel{2}} = 1275$

Q $1+2+3+4+\ldots+100 = \frac{\overset{50}{\cancel{100}} \times 101}{\cancel{2}} = \boxed{5050}$

Q $10+11+12+13\ldots\ldots+50$

ans = (1 से 50 तक) − (1 से 9 तक)

$= \frac{\overset{25}{\cancel{50}} \times 51}{\cancel{2}} - \frac{9 \times \overset{5}{\cancel{10}}}{\cancel{2}}$

$= 1275 - 45$

$= \boxed{1230}$

Q $9+10+11+\ldots+100$

Ans $1 \text{ से } 100 - 1 \text{ से } 8$

$$= \frac{100 \times 101}{2} - \frac{8 \times 9}{2}$$

$$= \frac{10100 - 72}{2}$$

$$= \frac{10028}{2}$$

$$5014$$

Natural No

- $1+2+3+4+\ldots+10 = \frac{n(n+1)}{2} = \frac{10\times 11}{2} = 55$
- $1+2+3+4+\ldots+50 = \text{''} = \frac{50\times 51}{2} = 1275$
- $1+2+3+4+\ldots+100 = \frac{100\times 101}{2} = 5050$
- $1^2+2^2+3^2+\ldots+10^2 = \frac{n(n+1)(2n+1)}{6} = \frac{10\times 11\times 21}{6} = 385$
- $1^2+2^2+3^2+\ldots+50^2 =$
- $1^3+2^3+3^3+\ldots+10^3 =$
- $1^3+2^3+3^3+\ldots+20^3 =$

① $1^2+2^2+3^2+\ldots+10^2 = \frac{n(n+1)(2n+1)}{6} = \frac{10\times11\times21}{6} = 385$

② $1^2+2^2+3^2+\ldots+50^2 = \frac{50\times51\times101}{6} = 42925$

③ $5^2+6^2+7^2+\ldots+11^2 =$

Ans

1 से 11 − 1 से 4

$= \frac{11\times12\times23}{6} - \frac{4\times5\times9}{6}$

$= 506 - 30$

$= 476$

Q $1^3+2^3+3^3+\ldots+8^3 = \left[\frac{n(n+1)}{2}\right]^2 = \left[\frac{8\times 9}{2}\right]^2 = 1296$ ✓

Q $1^3+2^3+3^3+\ldots+10^3 = \left[\frac{10\times 11}{2}\right]^2 = 3025$

Q $4^3+5^3+6^3+\ldots+9^3 =$

Ans 1^3 से 9 - 1^3 से 3

$= \left(\frac{9\times 10}{2}\right)^2 - \left(\frac{3\times 4}{2}\right)^2$

$= 2025 - 36$

$= 1989$

Q. $2+4+6+\ldots\ldots+20$

Ans

$$n = \frac{a_n - a}{d} + 1$$

$$n = \frac{20-2}{2} + 1 = 10$$

$$S_n = \frac{n}{2}(a + a_n)$$

$$= \frac{10 \times 22^{11}}{2}$$

$$= 110$$

$n = 10$

$$ans = n(n+1)$$

$$= 10 \times 11$$

$$= 110$$

Q $8+10+12+\ldots+50$

Ans $n = \frac{50-8}{2} + 1$

$= 22$

$S_n = \frac{11}{\cancel{22}} [8+50]$

$= 11 \times 58$

$= 638$

110. What is sum of odd numbers from 1 to 50?

1 से 50 तक विषम संख्याओं का योग क्या है?

(a) 625 **(b) 650**

(c) 667 **(d) 670**

$1+3+5+\ldots\ldots+49$

$n = \frac{a_n - a}{d} + 1$

$= \frac{49-1}{2} + 1 = 25$

$S_n = \frac{n}{2}[a + a_n]$

$= \frac{25}{2} \times 25 = 625$

1 से 50

25 odd | 25 even

n^2 | $n(n+1)$

$= 25^2$ | $= 25 \times 26$

$= 625$ | $= 650$

111. What is sum of first 50 odd numbers?

प्रथम 50 विषम संख्याओं का योग क्या है?

(a) 2500 **(b) 2600**

(c) 2700 **(d) 2800**

$n = 50$

$ans = n^2$

$= 50^2$

$= 2500$

$\frac{n(n+1)(2n+1)}{6}$

112. $7^2 + 8^2 + \ldots + 12^2 = ?$

(a) 459
(b) 559
(c) 567
(d) 570

1 से 12 − 1 से 6

$= \frac{12 \times 13 \times 25}{6} - \frac{6 \times 7 \times 13}{6}$

$= 650 - 91$

$= 559$

113. Find the value of $21^2 + 22^2 + 23^2 ..... + 30^2$

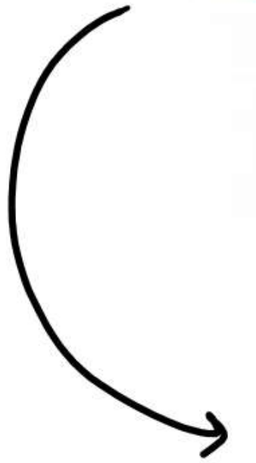
HW

$21^2 + 22^2 + 23^2 ..... + 30^2$ का मान ज्ञात कीजिए।

(a) 6855 (b) 6585

(c) 5865 (d) 8565

Q. $9^2 + 10^2 + \ldots + 14^2 + 15^2$



(1 से 15) − (1 से 8)

$= \frac{15 \times 16 \times 31}{6} - \frac{8 \times 9 \times 17}{6}$

$= 1240 - 204$

$= 1036$

$\left[\frac{n(n+1)}{2}\right]^2$

114. $9^3 + 10^3 + \ldots + 14^3 + 15^3$

(a) 12104 **(b) 12105**

(c) 13104 **(d) 14104**

1से 15 - 1से 8

$= \left(\frac{15 \times 16^{8}}{2}\right)^2 - \left(\frac{8^{4} \times 9}{2}\right)^2$

$= (120)^2 - (36)^2$

$= 14400 - 1296$

$= 13104$